BN

BUSINESSNow

DELIVERY COMMAND CENTER

Product Documentation

Comprehensive PSA Platform for Oracle Transportation Management Consulting Firms

11	30+	32	35+
User Roles	Modules	DB Tables	API Routes

Projects · Milestones · Tasks · Timesheets · Resources · Capacity · Finance · Invoices · Contracts · Rate Cards · Portfolio · Pipeline

KSAP Technologies
Oracle Transportation Management Consulting

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Confidential — Internal Use Only

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01 · Platform Overview & Architecture

BUSINESSNow — PSA Platform for KSAP Technologies

BUSINESSNow is a comprehensive Professional Services Automation (PSA) platform built specifically for KSAP Technologies, an Oracle Transportation Management (OTM) consulting firm. It provides an integrated command center for delivery operations, resource capacity, client health, and financial performance across all active engagements.

Frontend
React 19 + Vite 6, TailwindCSS, Radix UI, Recharts

Backend
Express 5 (TypeScript), REST API, role-based middleware

Database
PostgreSQL + Drizzle ORM (32 tables, full relational schema)

Runtime
Node 20 LTS, pnpm monorepo, KSAP-hosted infrastructure

Security
Rate limiting (100 req/15 min), requireRole RBAC middleware, morgan request logging, Sentry frontend monitoring

Ports
API :8080 · Web SPA :5173 · Preview proxy port-forwarding

PROJECT DELIVERY	PEOPLE & CAPACITY	CLIENT PIPELINE	FINANCE & CONTRACTS	EXECUTIVE & ADMIN
Projects	Team Roster	Customers	Financials / WIP	Portfolio View
Milestones	Utilization Heatmap	Customer Detail	Invoices	Admin Dashboard
Tasks	Assignments	Prospects	Rate Cards	System Admin Panel
Time Logs	Staffing Requests	Opportunity Pipeline	Change Orders	
Timesheet Approval	Capacity Forecast		Contract Manager	
Blueprints				

02 · User Roles & Permissions

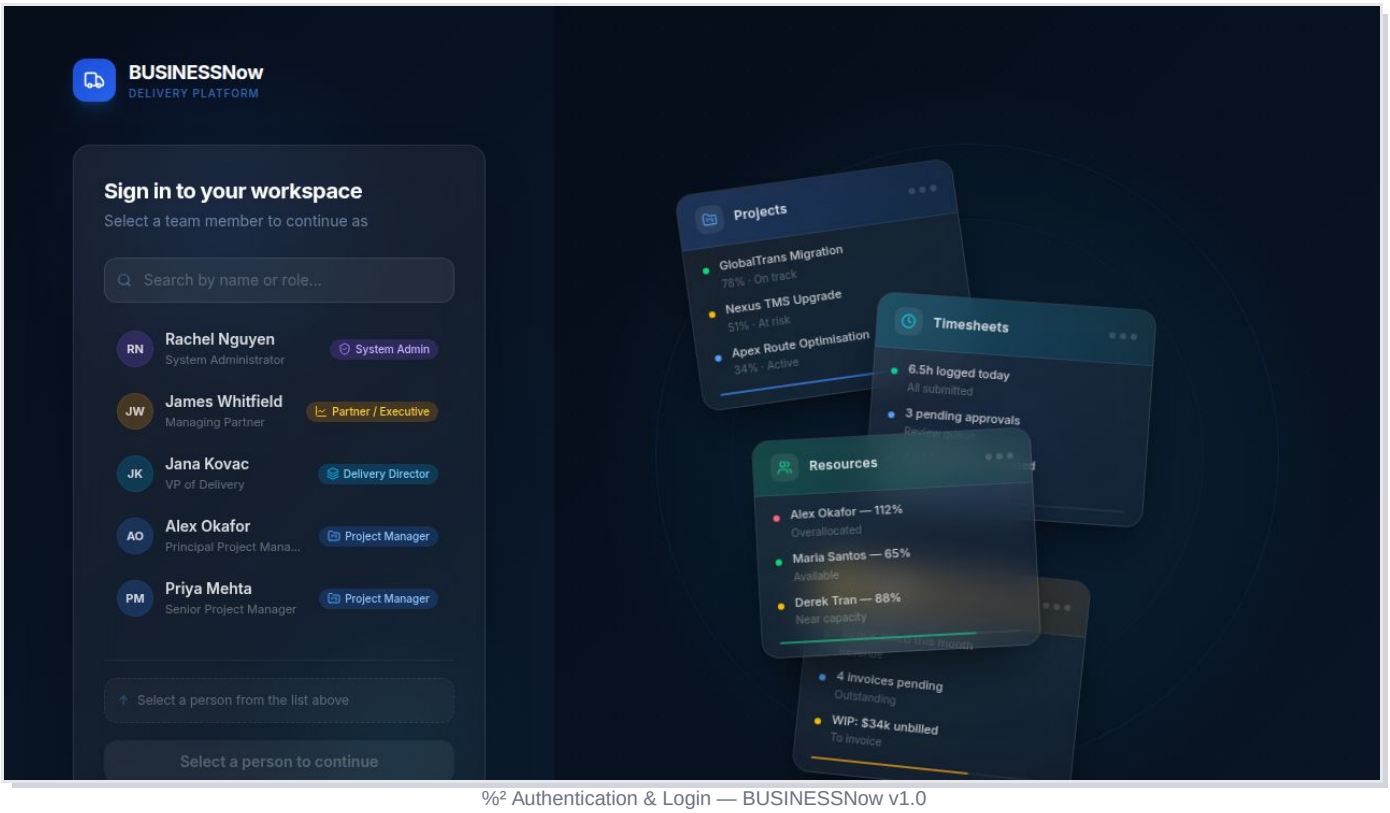
11 distinct roles with scoped access across all modules

System Admin key: admin	Full platform access. Manages users, system config, data integrity, audit logs.	All modules
Partner / Executive key: executive	Read-only portfolio view. KPIs, revenue, margin, at-risk projects. No editing.	Portfolio, Customers, Pipeline
Delivery Director key: delivery_director	Cross-project delivery oversight. Approves staffing, reviews all project health.	All delivery + resources
Project Manager key: project_manager	Manages assigned projects end-to-end. Creates milestones, tasks, approves timesheets.	Projects, team, finance (own)
Consultant key: consultant	Logs time, updates assigned tasks, views own project context.	Tasks, Time Logs, own projects
Resource Manager key: resource_manager	Manages staffing, allocations, capacity planning, bench reports.	Team, Assignments, Capacity
Finance Lead key: finance_lead	Full access to WIP, invoicing, rate cards, contracts, margin analytics.	All finance modules
Account Manager key: account_manager	Customer health, renewal tracking, executive relationship view.	Customers, Portfolio (read)
Sales / BD key: sales	Prospects pipeline, opportunity management, ACV tracking.	Prospects, Opportunities, Customers
PMO Analyst key: pmo	Templates, blueprints, process governance, cross-project reporting.	Templates, Portfolio (read)
Subcontractor key: subcontractor	Time entry only. Restricted to assigned project and task views.	Own Time Logs only

03 Authentication & Login

All roles

Role-based login screen. Team members select their identity from the roster. Each login loads a scoped workspace tailored to their permissions.



%² Authentication & Login — BUSINESSNow v1.0

KEY FEATURES

- User roster with avatar initials and role badges
- Role-based sidebar and module access on login
- Animated preview pane showing live platform metrics
- No password required in demo mode; JWT in production
- Auto-redirect to role-appropriate dashboard
- Session persisted in localStorage (otmnow_role)

REQUIREMENTS

REQ ID			
REQ-AUTH-01	Display all active users with role badge	Must	All
REQ-AUTH-02	Load role-scoped sidebar on login	Must	All
REQ-AUTH-03	Redirect to role dashboard after auth		

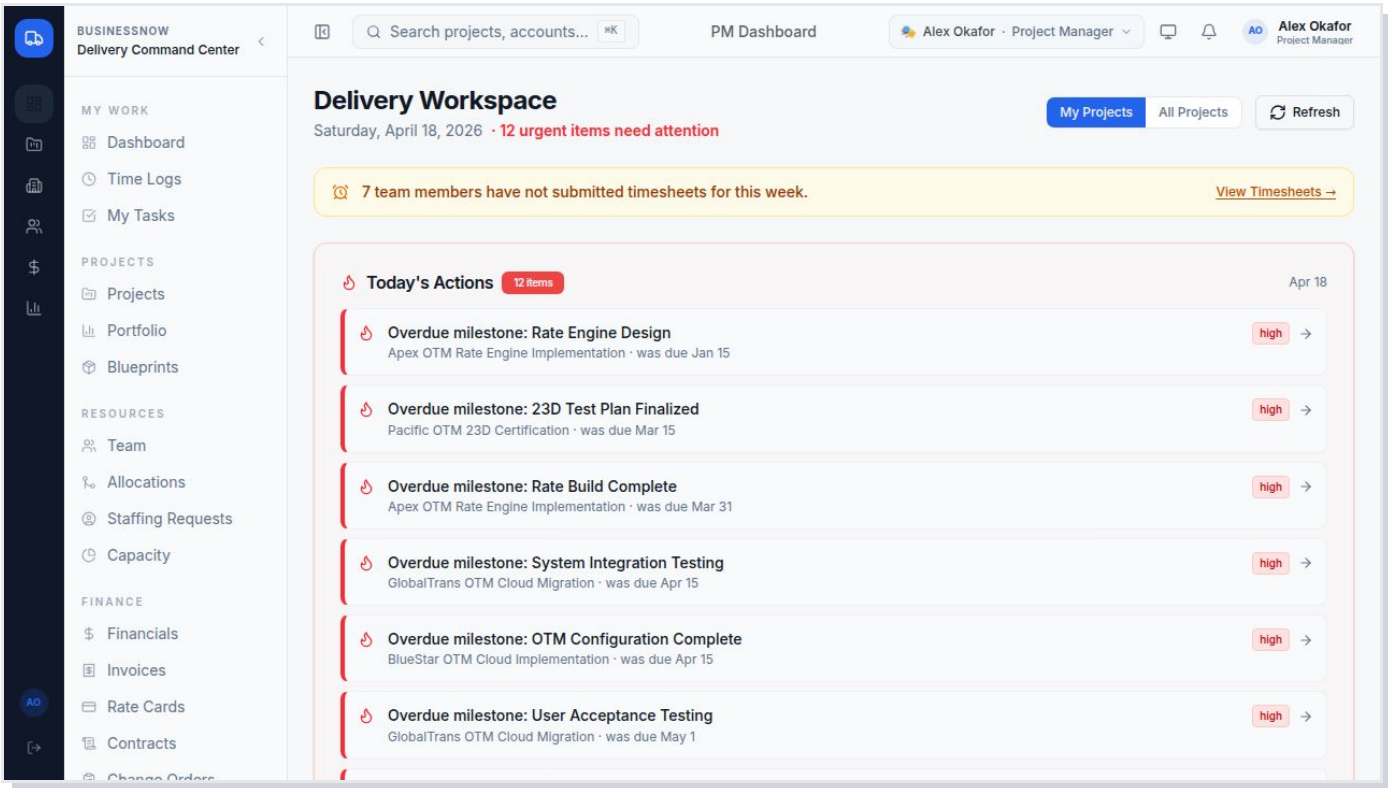
Must

All

04 PM Dashboard — Delivery Workspace

Project Manager

Personalized operational command center for Project Managers. Surfaces overdue milestones, urgent action items, and timesheet compliance alerts.



%² PM Dashboard — Delivery Workspace — BUSINESSNow v1.0

KEY FEATURES

- "Today's Actions" queue sorted by urgency and priority
- Overdue milestone alerts with project context and links
- Timesheet non-compliance warning for team members
- Real-time refresh button for live data updates
- Urgent item count badge on header
- Quick-navigate links to relevant module pages

REQUIREMENTS

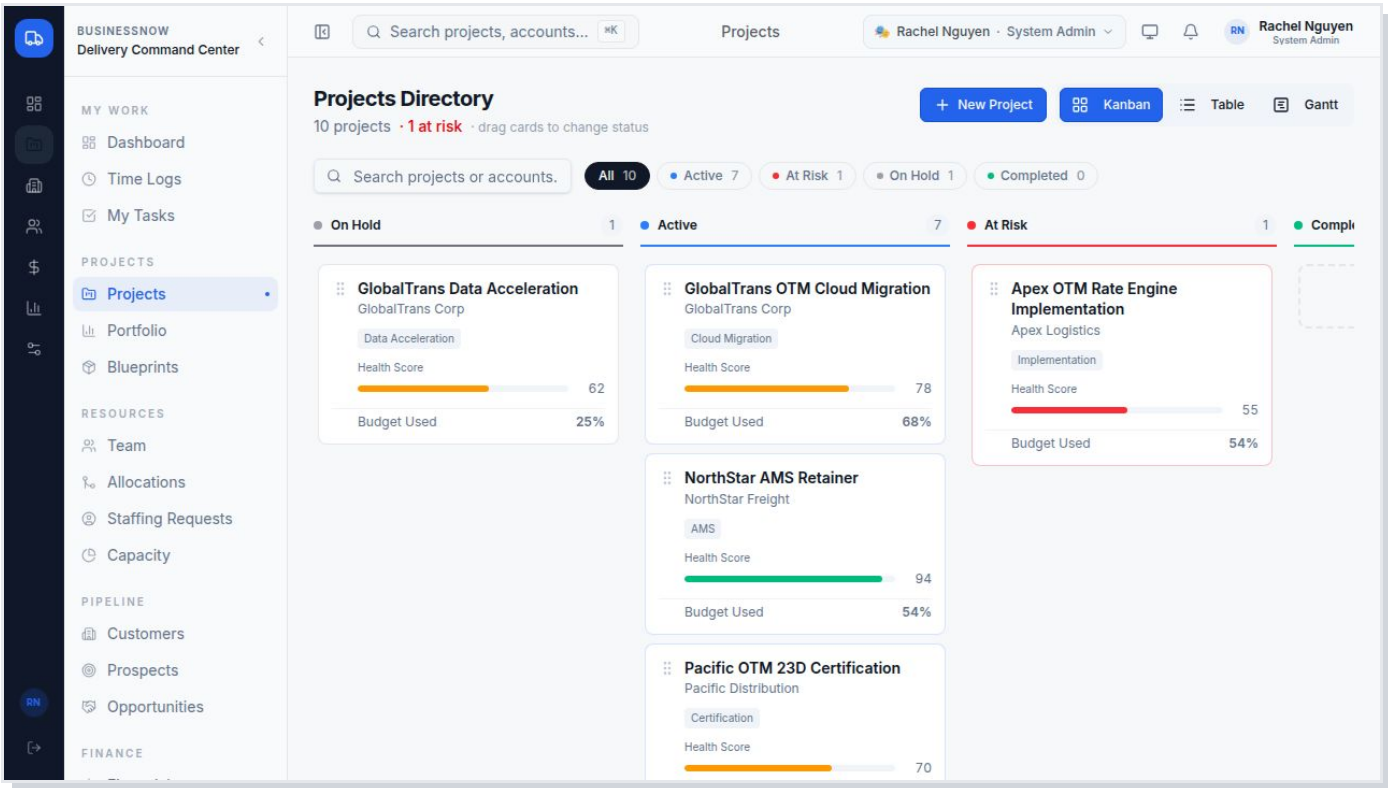
REQ ID			
REQ-DASH-01	Show urgent items count in header badge	Must	PM
REQ-DASH-02	Surface overdue milestones with due date & project	Must	PM
REQ-DASH-03	Alert on team members with missing		

	timesheets	Should	PM
REQ-DASH-04			
	One-click navigation to any action item	Must	PM

05 Projects — List View

Central registry of all client engagements. Shows health scores, budget burn, phase, and PM assignment. Supports kanban and table views.

PM · Delivery Director



%² Projects — List View — BUSINESSNow v1.0

KEY FEATURES

- Health score (0–100) with colour-coded bar indicator
- Budget burn % and total contract value
- Current phase, status, and PM assignment
- Filter by status, type, PM, customer
- Kanban view by delivery phase
- Quick-create new project from template

REQUIREMENTS

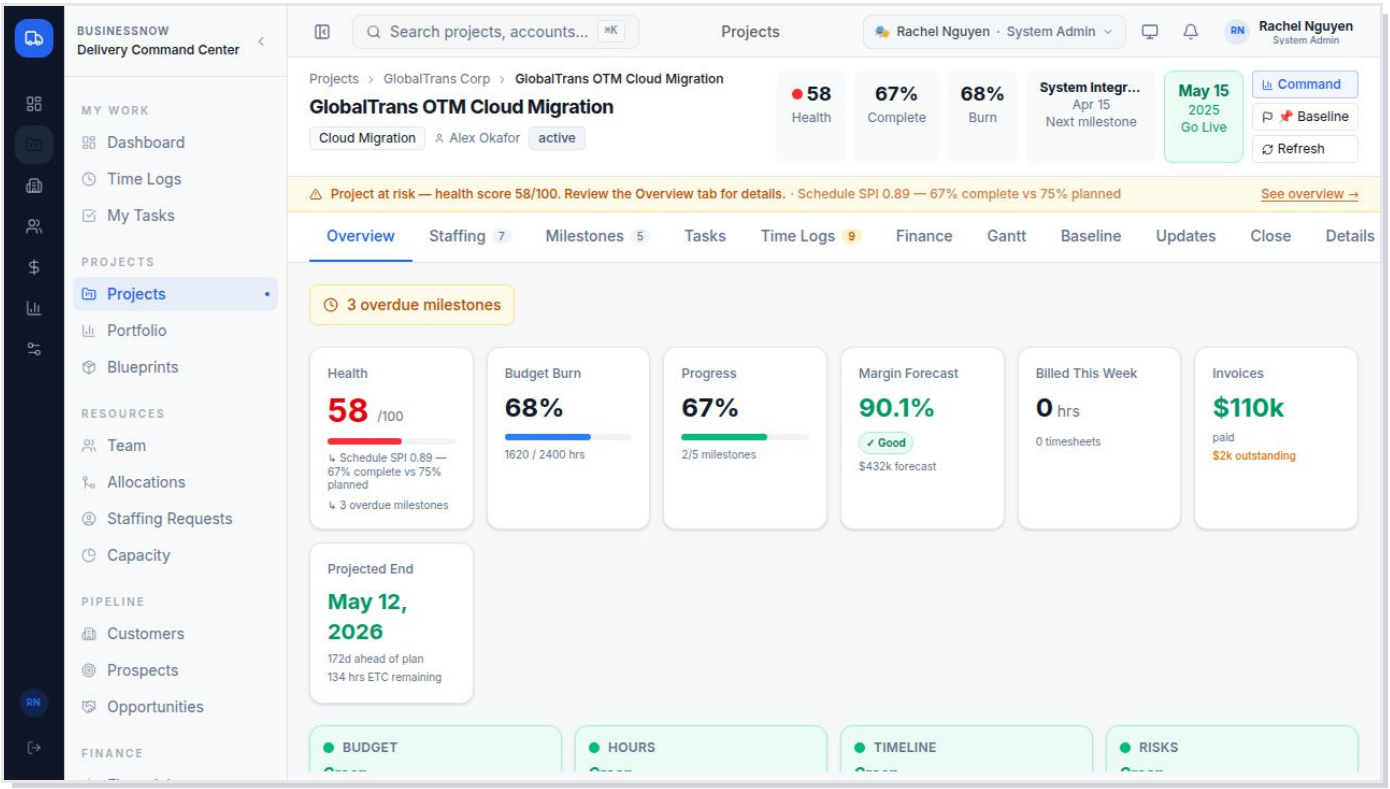
REQ ID			
REQ-PROJ-01	Display health score with visual indicator	Must	PM, DD
REQ-PROJ-02	Show budget burn and TCV per project	Must	PM, Finance
REQ-PROJ-03	Filter by status, type, PM, customer		

		Must	PM, DD
REQ-PROJ-04			
	Support Kanban and Table view toggle	Should	PM

06 Project Detail — Command Center

Project Manager

Deep-dive into a single engagement. Shows real-time health, budget burn, Gantt timeline, team roster, finance, and milestone status — all in one tabbed workspace.



%² Project Detail — Command Center — BUSINESSNow v1.0

KEY FEATURES

- Health score with Schedule Performance Index (SPI)
- Budget burn %, margin forecast, invoiced amount
- "Go Live" countdown with next milestone highlighted
- 9 tabs: Overview, Team, Milestones, Tasks, Time Logs, Finance, Gantt, Updates, Details
- At-risk banners with explanatory text
- Inline refresh and baseline comparison

REQUIREMENTS

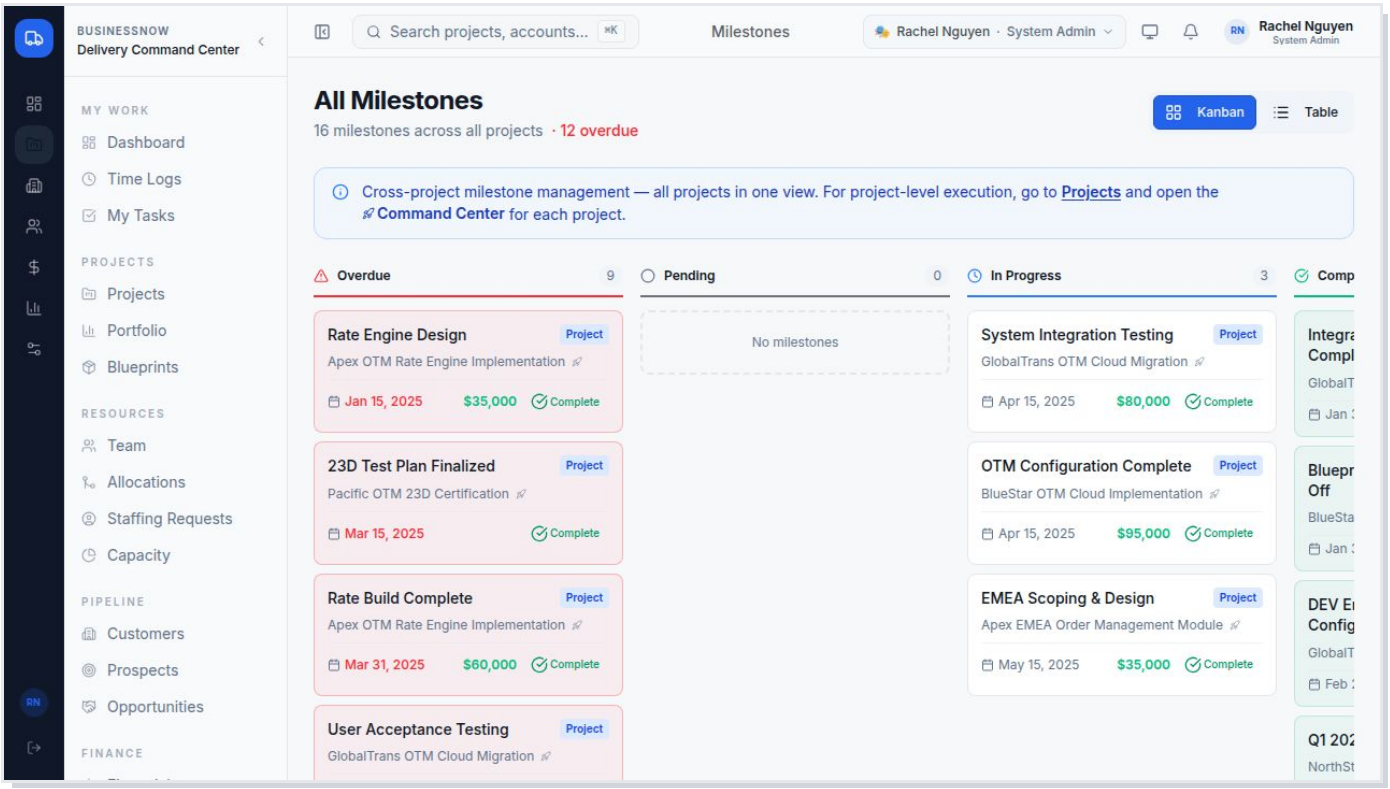
REQ ID			
REQ-PDET-01	Show SPI and CPI with risk bands	Must	PM
REQ-PDET-02	Tabbed interface with 9 contextual views	Must	PM
REQ-PDET-03			

	Display next Go-Live milestone prominently	Must	PM, DD
REQ-PDET-04			
	Margin forecast and revenue summary	Should	Finance

07 Milestones — Cross-Project Tracking

Project Manager

All delivery milestones across every active engagement in a single Kanban board. Tracks status, payment triggers, and overdue items at a glance.



%² Milestones — Cross-Project Tracking — BUSINESSNow v1.0

KEY FEATURES

- Columns: Overdue · Pending · In Progress · Complete
- Colour-coded status with payment amount per milestone
- Cross-project aggregation with project name context
- Table view alternative with sort & filter
- Overdue count badge per column
- Direct link to parent project command center

REQUIREMENTS

REQ ID			
REQ-MILE-01	Kanban columns by milestone status	Must	PM, DD
REQ-MILE-02	Show payment trigger amount per milestone	Must	Finance
REQ-MILE-03	Overdue highlighting with due date		

Must

PM

REQ-MILE-04

Cross-project view aggregating all milestones

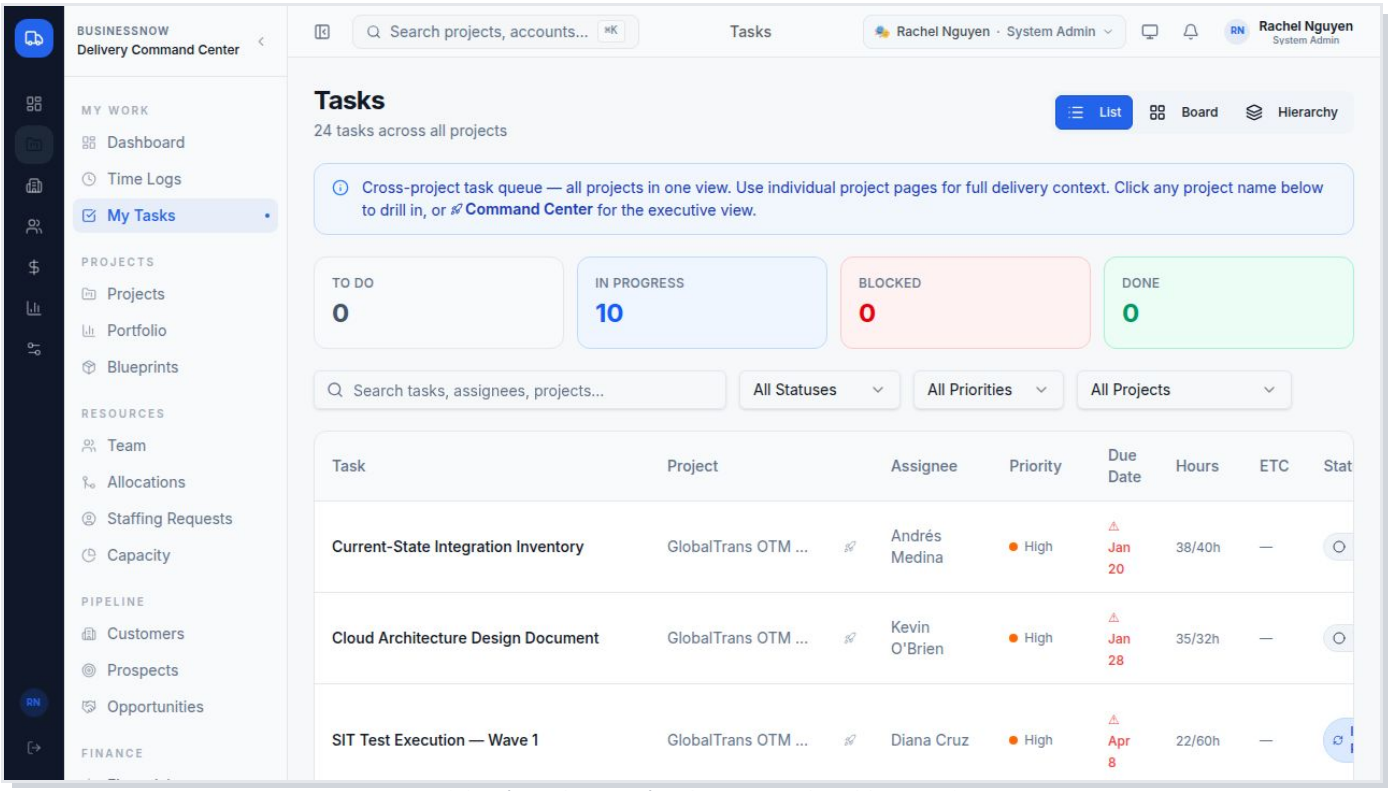
Must

DD

08 Tasks — Cross-Project Queue

Aggregated task list across all projects. Summary cards for TO DO, IN PROGRESS, BLOCKED, and DONE. Supports list, board, and hierarchy (WBS) views.

Project Manager · Consultant



%² Tasks — Cross-Project Queue — BUSINESSNow v1.0

KEY FEATURES

- Status summary cards with count per state
- Assignee, priority, due date, hours logged, ETC columns
- Filter by status, priority, project, assignee
- Board view (Kanban) and Hierarchy (WBS) modes
- Search across task name, assignee, project
- Inline status update with dropdown

REQUIREMENTS

REQ ID			
REQ-TASK-01	Status summary cards (TO DO, IN PROGRESS, BLOCKED, DONE)	Must	PM, Consultant
REQ-TASK-02	Filter by status, priority, project, assignee	Must	PM
REQ-TASK-03	List, Board, and Hierarchy (WBS) view modes		

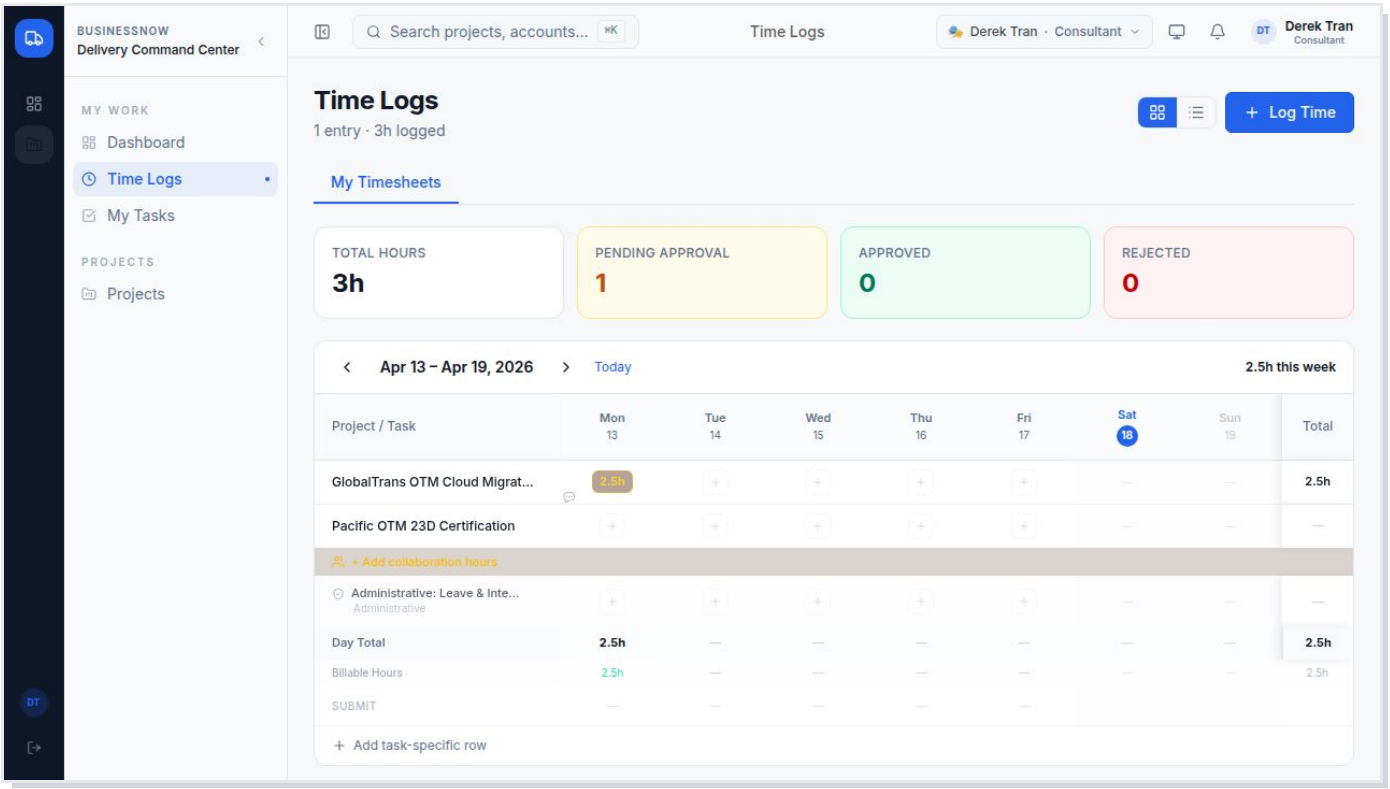
Should

PM

09 Time Logs — Weekly Timesheet

Weekly time-entry grid with billable/non-billable split. Consultants log hours against project-task rows. Shows totals, submission status, and approval state.

Consultant · All billing roles



%² Time Logs — Weekly Timesheet — BUSINESSNow v1.0

KEY FEATURES

- Mon–Sun grid per project/task combination
- Billable vs. total hours footer per day
- TOTAL HOURS, PENDING, APPROVED, REJECTED summary cards
- Collaboration hours row for internal team time
- Week navigation with today highlight
- Submit button per week with status rollup

REQUIREMENTS

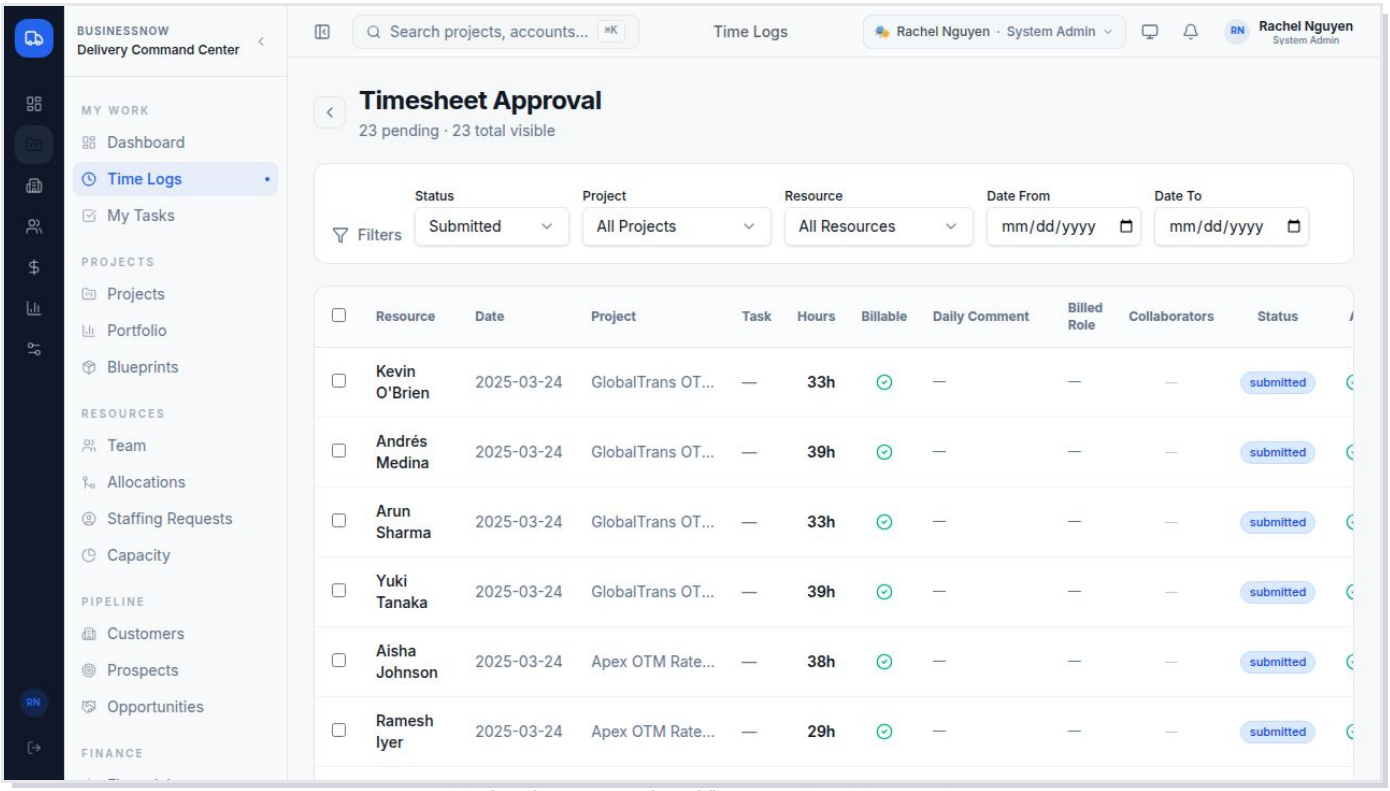
REQ ID			
REQ-TIME-01	Weekly grid with Mon–Sun columns per project/task	Must	Consultant
REQ-TIME-02	Separate billable vs non-billable tracking	Must	Finance
REQ-TIME-03	Submit timesheet week for PM approval		

		Must	Consultant
REQ-TIME-04			
	Status summary cards (Pending, Approved, Rejected)	Must	Consultant

10 Timesheet Approval Workflow

Project Manager

PM-facing queue of all submitted timesheets. Bulk review with approve / reject actions per entry. Filters by status, project, resource, and date range.



%² Timesheet Approval Workflow — BUSINESSNow v1.0

KEY FEATURES

- All submitted timesheets in a tabular queue
- 23 pending approvals shown with resource, project, hours
- Billable flag and billed-role assignment per entry
- Approve (') / Reject (') action buttons inline
- Filter by project, resource, date range
- Bulk select with checkbox column for mass approval

REQUIREMENTS

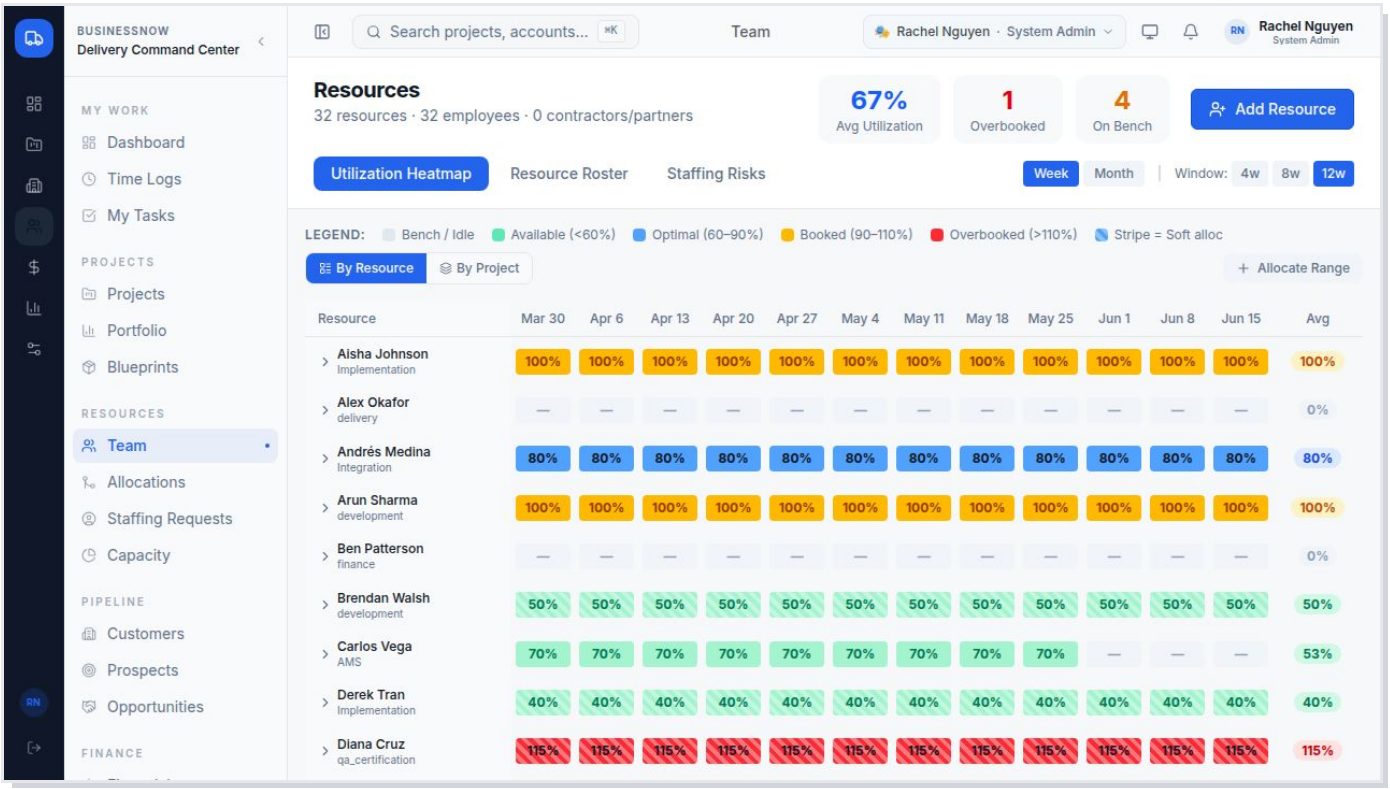
REQ ID			
REQ-TAPPR-01	Queue of all submitted timesheets for PM review	Must	PM
REQ-TAPPR-02	Inline approve and reject actions with icon buttons	Must	PM
REQ-TAPPR-03			

	Filter by project, resource, date range, status	Must	PM
REQ-TAPPR-04			
	Bulk select and mass approve functionality	Should	PM

11 Team — Resource Utilization Heatmap

Resource Manager

12-week rolling utilization heatmap for all 32 team members. Colour-coded by utilization band. Toggles between "By Resource" and "By Project" pivot views.



%² Team — Resource Utilization Heatmap — BUSINESSNow v1.0

KEY FEATURES

- Colour legend: Bench · Available · Optimal · Booked · Overbooked
- 12-week rolling window (4w / 8w / 12w selector)
- Average utilization % in rightmost column
- Click row to expand individual resource timeline
- "By Project" view shows same heatmap pivoted to projects
- Overbooked / On-bench counts in header KPI chips

REQUIREMENTS

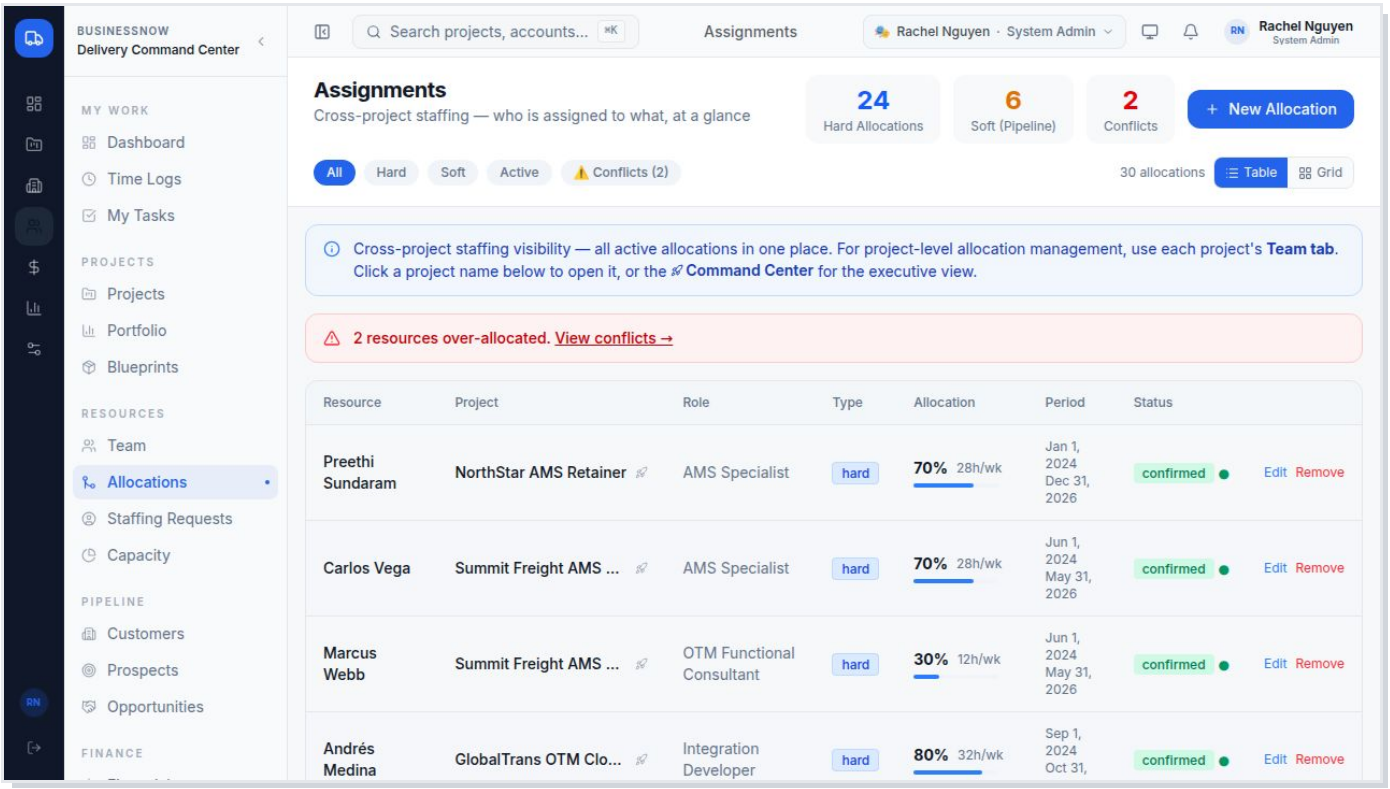
REQ ID			
REQ-TEAM-01	Heatmap with utilization colour bands per resource/week	Must	RM
REQ-TEAM-02	4w / 8w / 12w window selector	Must	RM
REQ-TEAM-03	By Resource and By Project pivot toggle		

		Should	RM
REQ-TEAM-04			
	Overbooked count with drill-down	Must	RM

12 Assignments — Cross-Project Staffing

Resource Manager

All active resource allocations in one table. Shows hard vs. soft allocations, conflict detection, hours/week, and confirmation status.



%² Assignments — Cross-Project Staffing — BUSINESSNow v1.0

KEY FEATURES

- 24 hard allocations, 6 soft (pipeline), 2 conflicts shown
- Allocation type: Hard (confirmed) · Soft (pipeline)
- % utilization with hours/week display
- Date range with start/end period per allocation
- Conflict banner: "2 resources over-allocated"
- New Allocation quick-create button

REQUIREMENTS

REQ ID			
REQ-ALLOC-01	Display all allocations with hard/soft type	Must	RM
REQ-ALLOC-02	Detect and surface over-allocation conflicts	Must	RM
REQ-ALLOC-03	Filter by All, Hard, Soft, Active, Conflicts		

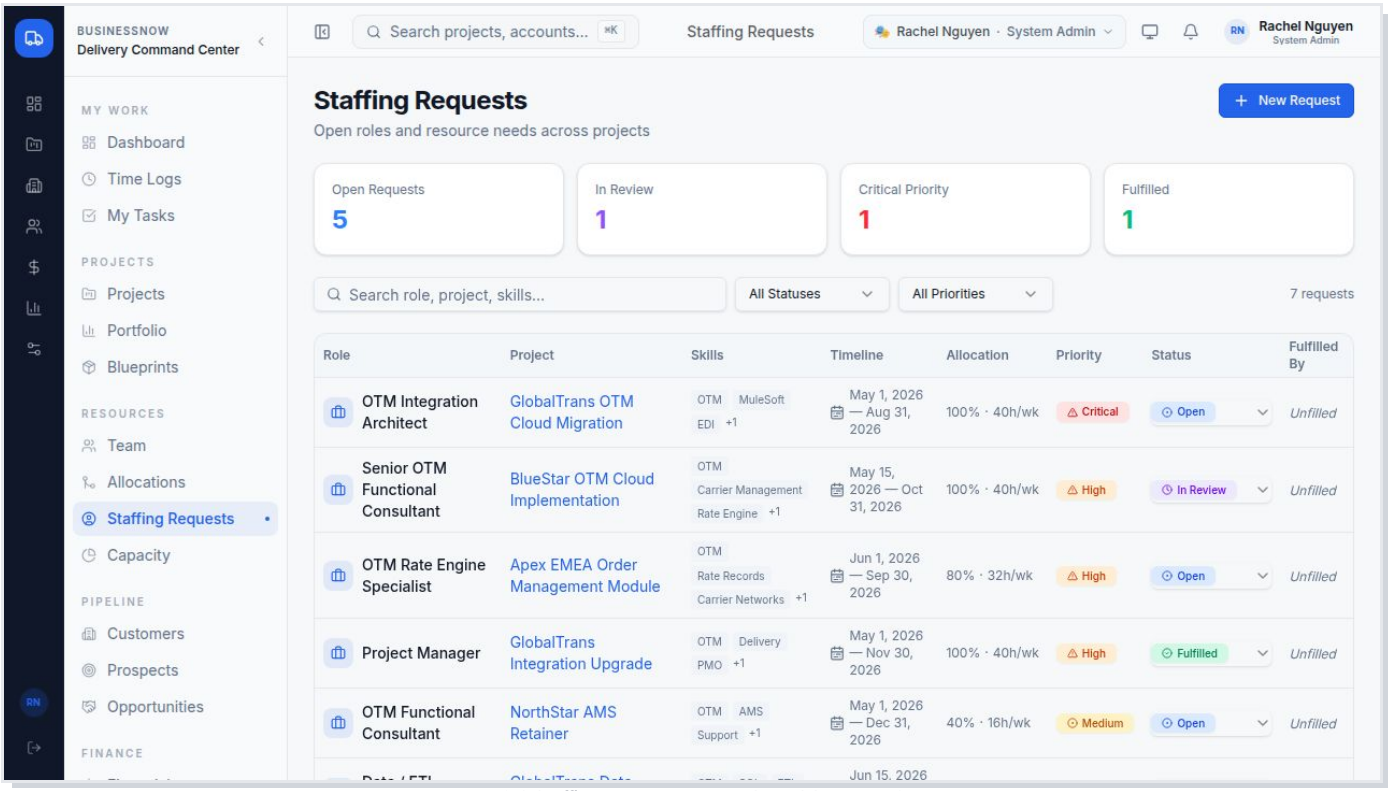
Must

RM

13 Staffing Requests

Open role requisitions across all projects. Tracks critical and high-priority staffing gaps with required skills, timeline, and allocation percentage.

Resource Manager · PM



%² Staffing Requests — BUSINESSNow v1.0

KEY FEATURES

- 5 open, 1 in review, 1 critical priority, 1 fulfilled
- Role, skills, project, timeline, allocation, and priority columns
- Critical / High / Medium / Low priority badge system
- Status dropdown: Open · In Review · Fulfilled · Cancelled
- Linked to parent project for context
- New Request CTA with role and skills fields

REQUIREMENTS

REQ ID			
REQ-SREQ-01	List open staffing requests with priority badge	Must	RM, PM
REQ-SREQ-02	Skills, timeline, allocation %, project context	Must	RM
REQ-SREQ-03	Status workflow: Open !’ In Review !’ Fulfilled		

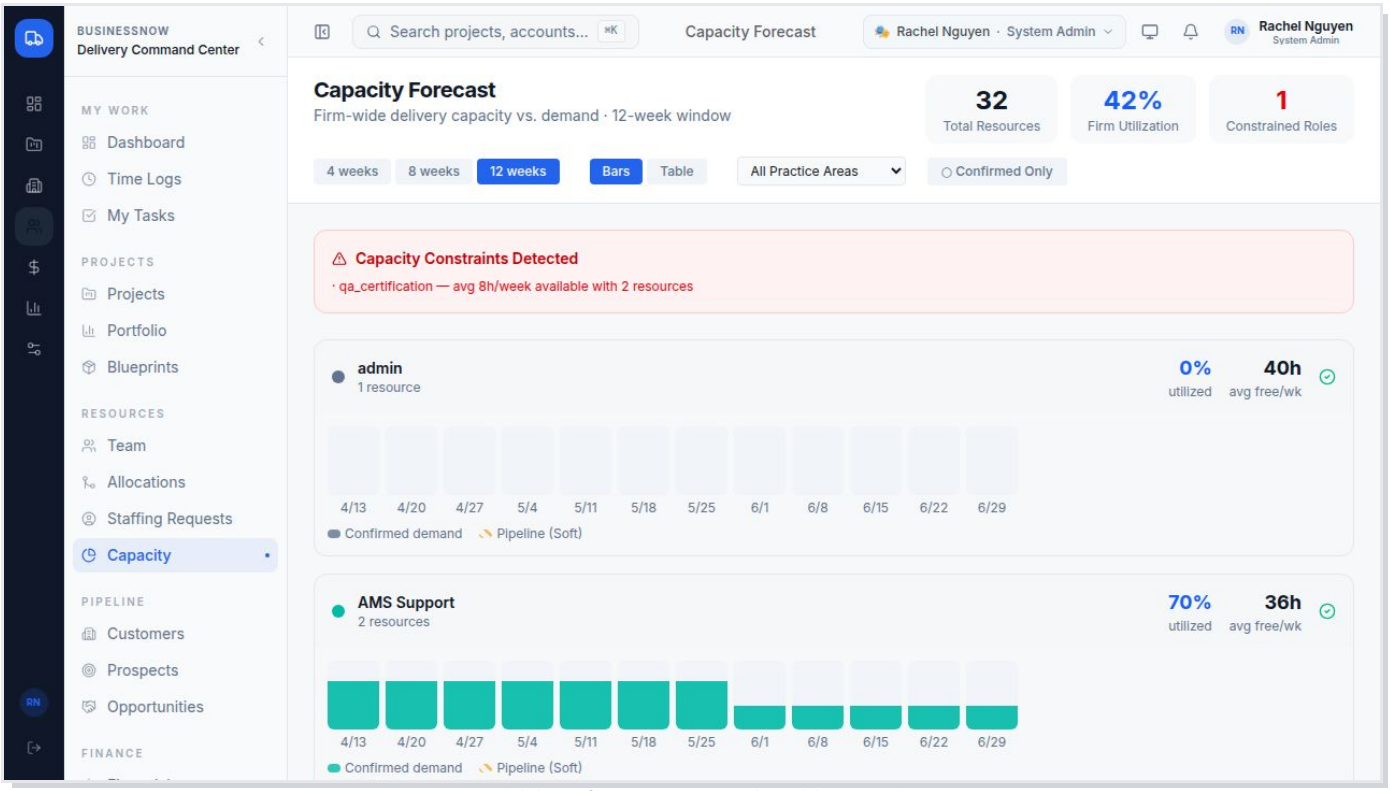
Must

RM

14 Capacity Forecast

Resource Manager

12-week bar chart forecast of hard demand vs. soft pipeline demand per OTM specialization. Surfaces constrained roles and capacity gaps.



%² Capacity Forecast — BUSINESSNow v1.0

KEY FEATURES

- 32 resources across all specializations tracked
- 12-week window with 4w/8w options
- Hard demand (solid) vs. Soft demand (pipeline) per bar
- Capacity constraint banner for bottleneck roles
- Grouped by OTM specialization (AMS, Implementation, etc.)
- Bars toggle: weekly bars vs. table view

REQUIREMENTS

REQ ID			
REQ-CAP-01	Bar chart of capacity vs demand per specialization	Must	RM, DD
REQ-CAP-02	Hard demand vs soft pipeline differentiation	Must	RM
REQ-CAP-03	Capacity constraint alerts for bottleneck roles		

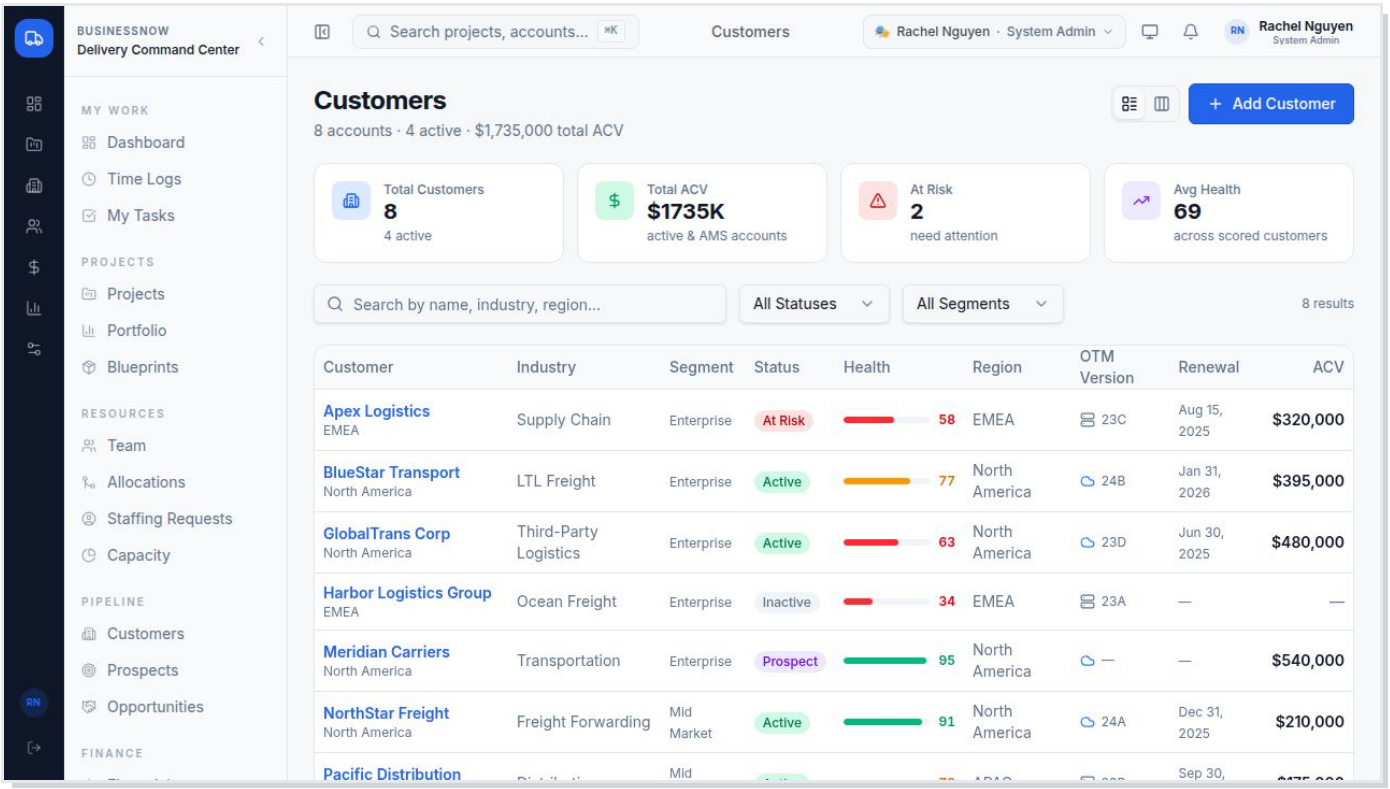
Must

RM, DD

15 Customers (Accounts)

8 enterprise accounts with health scoring, OTM version tracking, ACV, renewal dates, region, and segment. Two at-risk accounts highlighted.

Account Manager · PM



%² Customers (Accounts) — BUSINESSNow v1.0

KEY FEATURES

- 8 accounts: 4 active, 4 prospect/inactive
- Health score 0–100 with visual progress bar
- OTM version per account (23A, 23C, 23D, 24A, 24B)
- ACV, renewal date, industry, and region columns
- "At Risk" badge for health < 65
- Filter by status and segment (Enterprise / Mid-Market)

REQUIREMENTS

REQ ID			
REQ-CUST-01	Health score with visual bar per customer	Must	AM, PM
REQ-CUST-02	OTM version, ACV, renewal date displayed	Must	AM
REQ-CUST-03	At-risk flag for health score < 65		

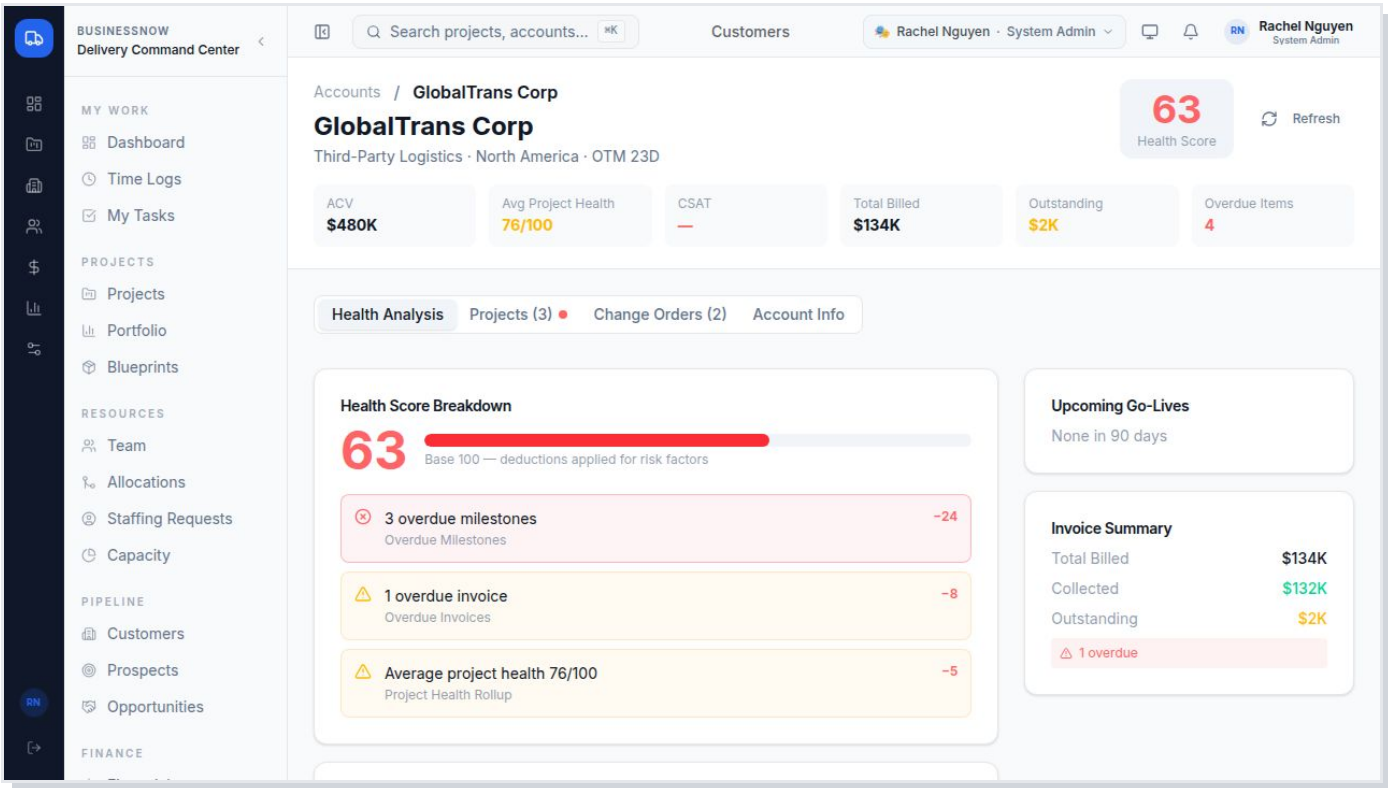
Must

AM, DD

16 Customer Detail — Health & Invoice Summary

Account Manager

Deep-dive account view with health score breakdown, deduction factors, invoice summary, project roster, change order history, and account info tabs.



%² Customer Detail — Health & Invoice Summary — BUSINESSNow v1.0

KEY FEATURES

- Health Score: 63/100 with deduction breakdown
- Deductions: -24 overdue milestones, -8 overdue invoice, -5 avg health
- Upcoming Go-Lives and invoice collection summary
- 4 tabs: Health Analysis · Projects · Change Orders · Account Info
- Outstanding and collected revenue with overdue flag
- Project list with individual health scores

REQUIREMENTS

REQ ID			
REQ-CDET-01	Health score breakdown with deduction reasons	Must	AM, DD
REQ-CDET-02	Invoice summary with outstanding / overdue amounts	Must	Finance, AM
REQ-CDET-03			

Tabbed view: Health, Projects, Change Orders, Info

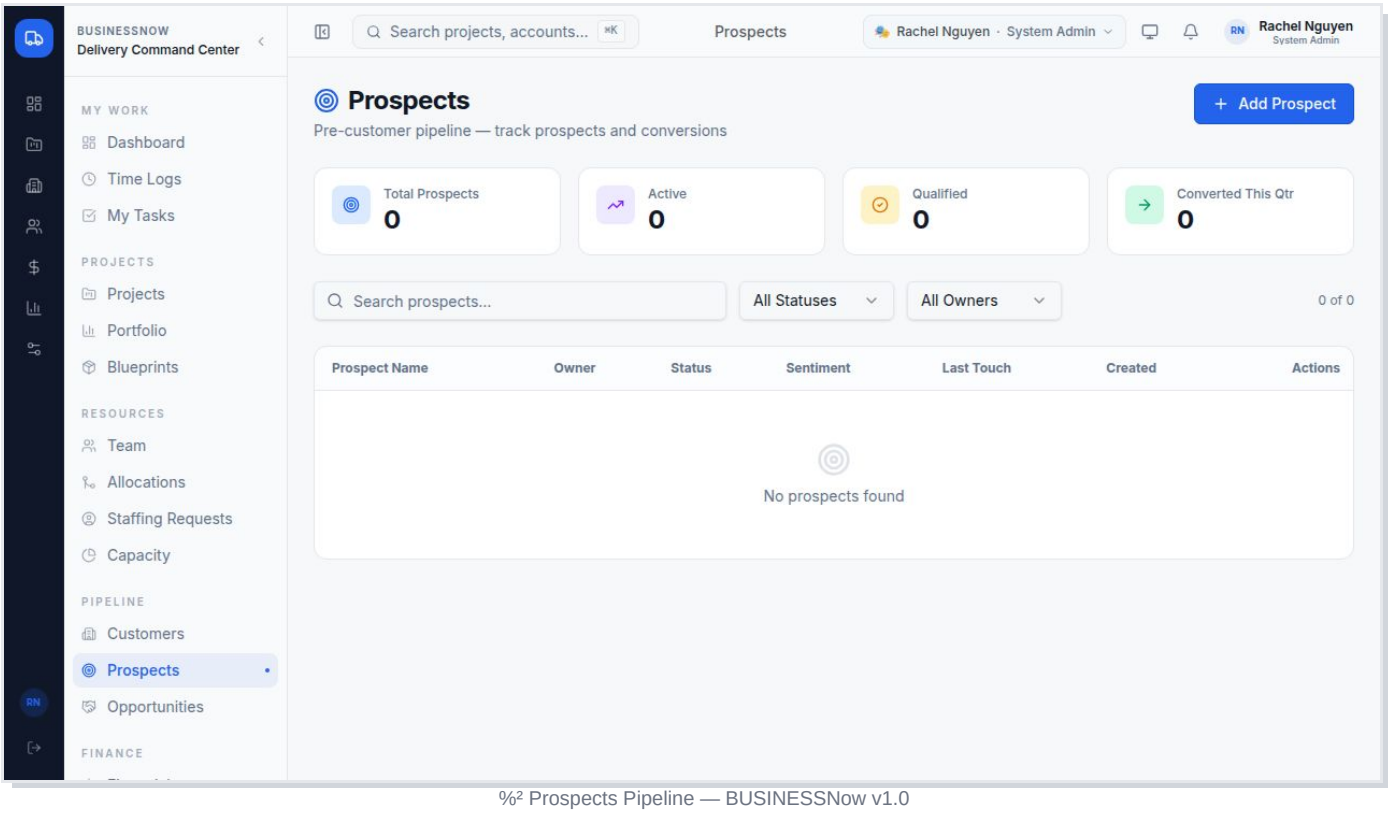
Must

AM

17 Prospects Pipeline

Prospects are managed within the Customers module, filtered by Prospect status. Meridian Carriers ($540K ACV) is shown as a prospect account pending qualification.

Sales · Account Manager



%² Prospects Pipeline — BUSINESSNow v1.0

KEY FEATURES

- Prospect accounts shown with "Prospect" status badge
- ACV potential, industry, region, segment visible
- Health score not yet calculated (pre-engagement)
- Qualification status and OTM version TBD
- Convert to Customer action when deal closes
- Link to Opportunity record for sales context

REQUIREMENTS

REQ ID			
REQ-PROS-01	Prospect accounts visible with status badge	Must	Sales, AM
REQ-PROS-02	ACV, region, industry, segment shown	Must	Sales
REQ-PROS-03	Convert prospect to customer on deal close		

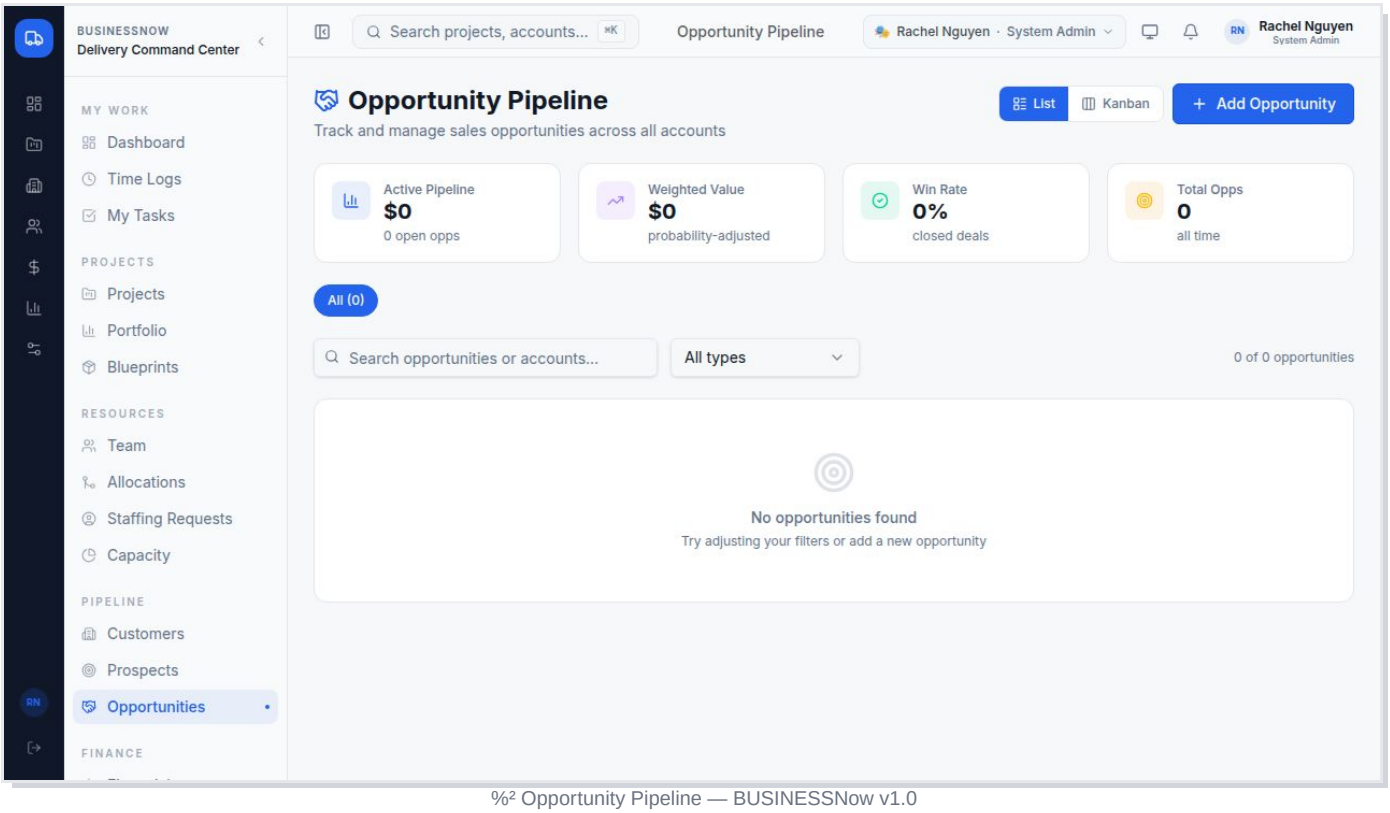
Should

Sales, AM

18 Opportunity Pipeline

Sales · Business Development

Full-lifecycle opportunity tracking. 10 opportunities across 7 stages from Lead through Won/Lost. $1.7M active pipeline, $873K weighted value, 67% win rate.



%² Opportunity Pipeline — BUSINESSNow v1.0

KEY FEATURES

- $1.7M active pipeline with 7 open opportunities
- Probability-adjusted weighted value: $873K
- Stage tabs: Lead · Qualified · Discovery · Proposal · Negotiation · Won · Lost
- Value, probability %, close date, owner per row
- Opportunity type: Cloud Migration, AMS, Custom Dev, Certification
- Add Opportunity with guided form

REQUIREMENTS

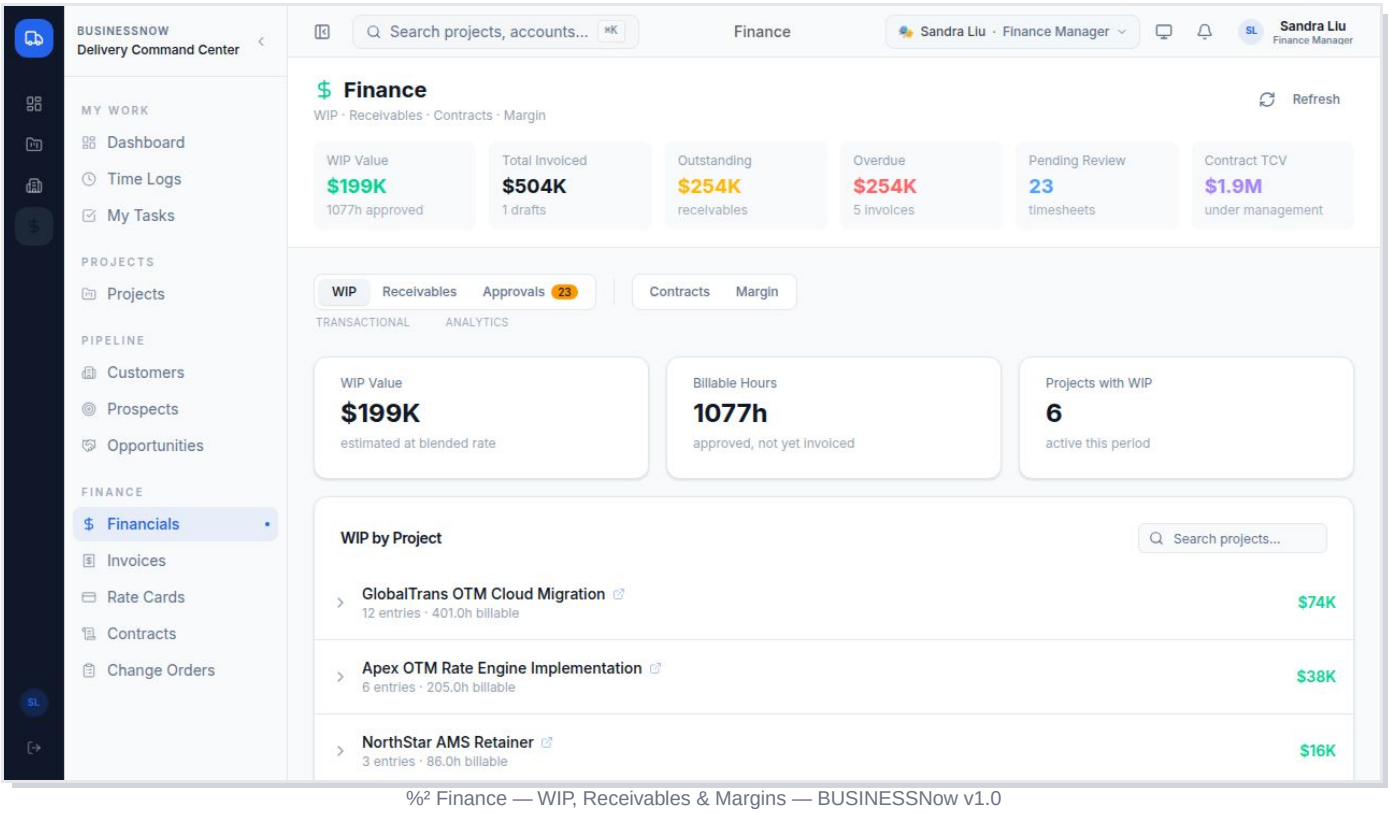
REQ ID			
REQ-OPP-01	Pipeline total and weighted value KPI cards	Must	Sales, DD
REQ-OPP-02	Stage-based pipeline with tab filters	Must	Sales
REQ-OPP-03			

	Probability %, value, close date, owner per opp	Must	Sales
REQ-OPP-04			
	Kanban view by stage	Should	Sales

19 Finance — WIP, Receivables & Margins

Finance Lead

Unified financial dashboard. WIP value, total invoiced, outstanding receivables, overdue invoices, timesheet approvals pending, and total contract value — all in one pane.



%² Finance — WIP, Receivables & Margins — BUSINESSNow v1.0

KEY FEATURES

- WIP Value: $199K (1077h approved, not yet invoiced)
- Total Invoiced: $672K (1 draft pending)
- Outstanding Receivables: $422K across 5 invoices
- 23 timesheets pending review and approval
- Contract TCV: $1.9M under management
- Tabs: WIP · Receivables · Approvals · Contracts · Margin

REQUIREMENTS

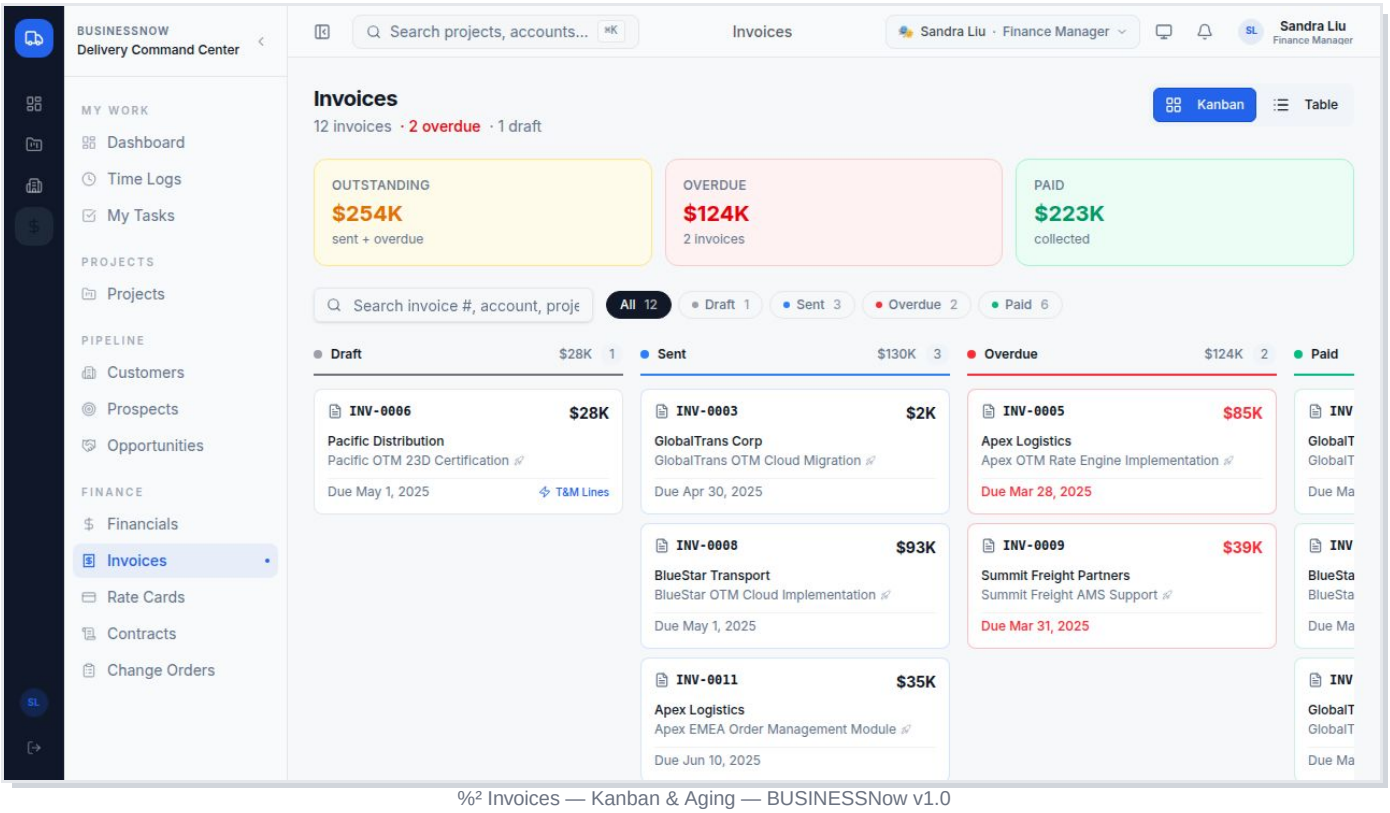
REQ ID			
REQ-FIN-01	WIP value, hours, and project breakdown	Must	Finance
REQ-FIN-02	Outstanding receivables and overdue count	Must	Finance
REQ-FIN-03	Pending timesheet approval count with quick link		

		Must	Finance, PM
REQ-FIN-04			
	Contract TCV and margin analytics tabs	Should	Finance, DD

20 Invoices — Kanban & Aging

Finance Lead

12 invoices in a 4-column Kanban: Draft · Sent · Overdue · Paid. $422K outstanding, $124K overdue across 2 invoices. Status filter tabs and invoice search.



%² Invoices — Kanban & Aging — BUSINESSNow v1.0

KEY FEATURES

- 4-column Kanban: Draft ($28K) · Sent ($298K) · Overdue ($124K) · Paid
- Invoice card: number, account, project, amount, due date
- Overdue invoices highlighted in red with past-due date
- Table view alternative with sorting
- Status filter tabs: All · Draft · Sent · Overdue · Paid
- Search by invoice #, account, or project name

REQUIREMENTS

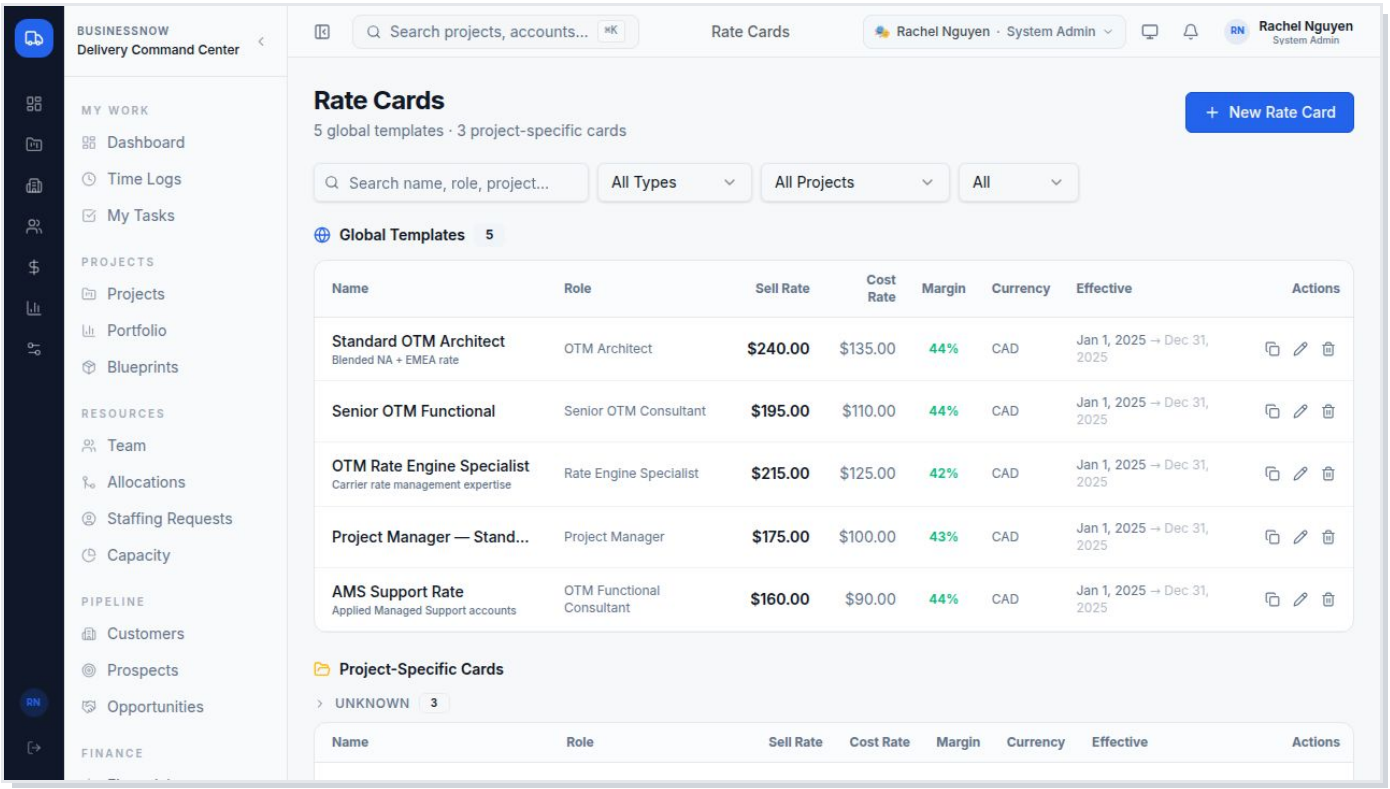
REQ ID			
REQ-INV-01	Kanban view: Draft, Sent, Overdue, Paid columns	Must	Finance
REQ-INV-02	Overdue invoice highlighting with past-due date	Must	Finance
REQ-INV-03			

	Invoice totals per column in header	Must	Finance
REQ-INV-04			
	Table view with sort and search	Should	Finance

21 Rate Cards

Finance Lead

5 global billing rate templates plus 3 project-specific overrides. Shows sell rate, cost rate, and margin % per OTM role. Used to calculate WIP and invoice amounts.



%² Rate Cards — BUSINESSNow v1.0

KEY FEATURES

- 5 global templates: OTM Architect, Senior Functional, Rate Engine Specialist, PM, AMS Support
- Sell rate and cost rate with margin % (42–44%) displayed
- Project-specific cards override global rates per engagement
- Currency: CAD with effective date range
- New Rate Card creation with role and rate fields
- Used by WIP engine to calculate billable revenue

REQUIREMENTS

REQ ID			
REQ-RATE-01	Global templates and project-specific card separation	Must	Finance
REQ-RATE-02	Sell rate, cost rate, and margin % per role	Must	Finance
REQ-RATE-03	Effective date range for rate validity		

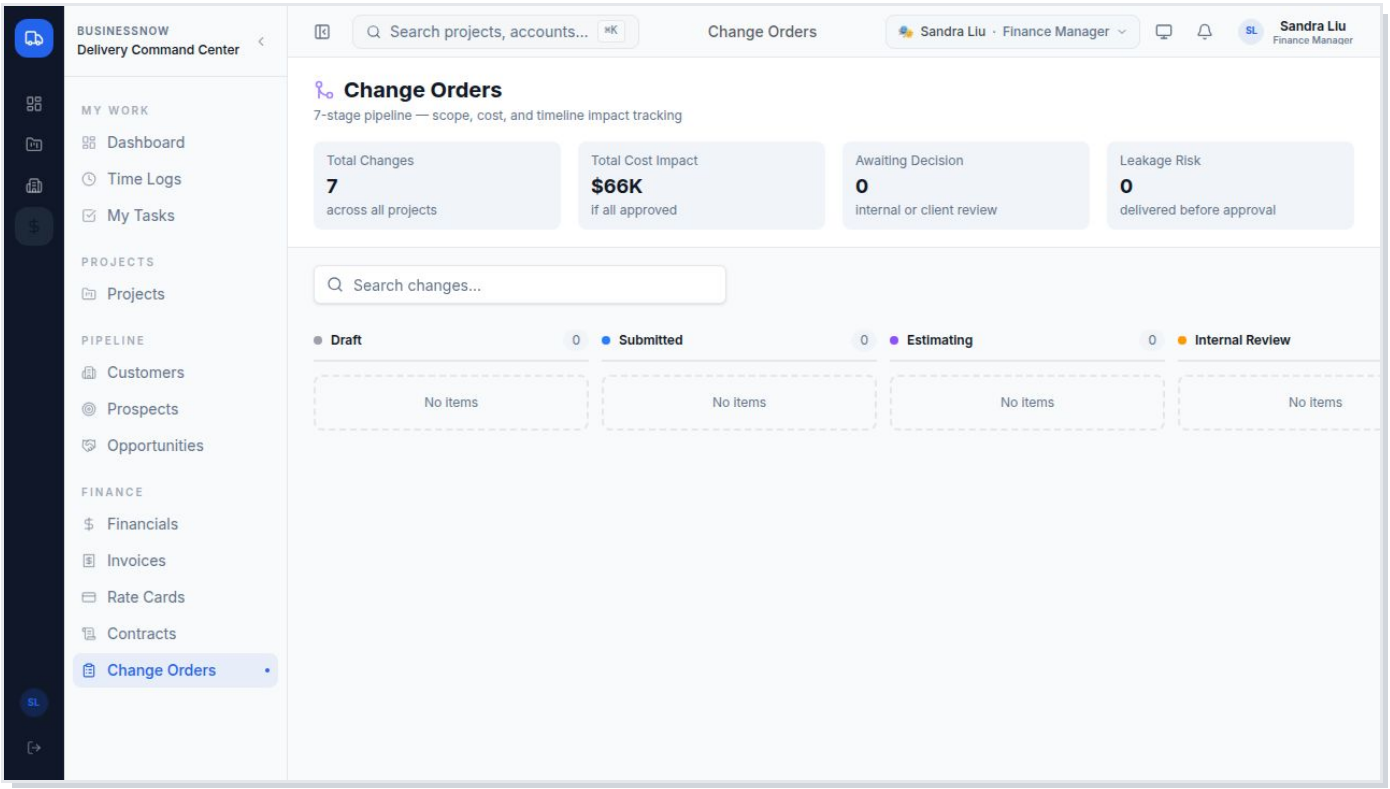
Must

Finance

22 Change Orders

7-stage change order pipeline tracking scope, cost, and timeline impacts. $66K total cost impact across 7 change orders if all approved.

Finance Lead · PM



%² Change Orders — BUSINESSNow v1.0

KEY FEATURES

- 7 change orders across all active projects
- Total cost impact: $66K if all approved
- 7-stage Kanban: Draft · Submitted · Estimating · Internal Review · Client Review · Approved · Rejected
- Awaiting Decision: 0 (none in client review currently)
- Leakage Risk: 0 (no delivered-before-approval items)
- Change type, scope impact, timeline delta per card

REQUIREMENTS

REQ ID			
REQ-CHG-01	7-stage Kanban pipeline for change orders	Must	Finance, PM
REQ-CHG-02	Total cost impact and count KPI cards	Must	Finance
REQ-CHG-03	Leakage risk detection (delivered before approval)		

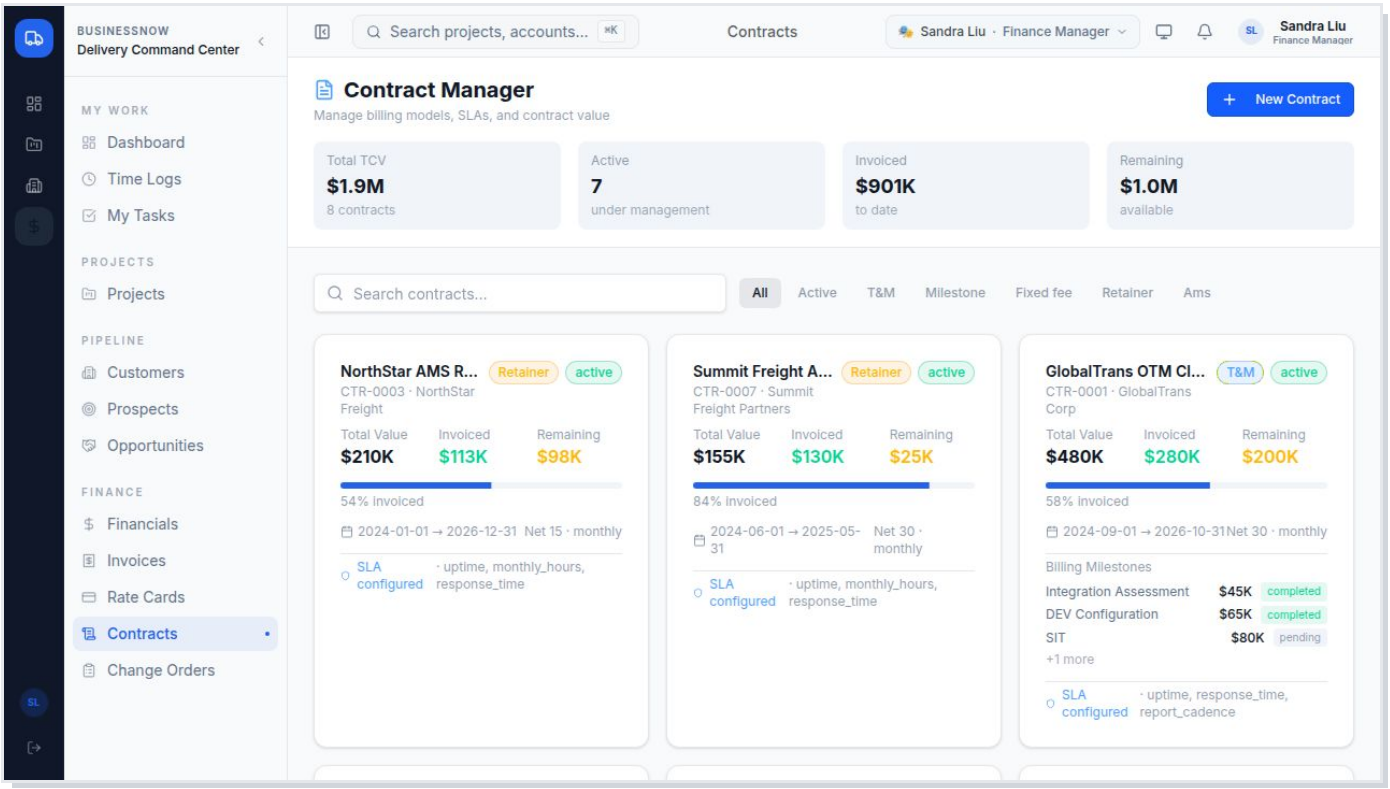
Must

Finance, DD

23 Contract Manager

Finance Lead

8 contracts totalling $1.9M TCV. Tracks billing model (T&M, Fixed Fee, Retainer, Milestone), invoiced amount, remaining value, SLA config, and payment terms.



%² Contract Manager — BUSINESSNow v1.0

KEY FEATURES

- 8 contracts: $1.9M TCV, $901K invoiced, $1.0M remaining
- Billing models: T&M · Milestone · Fixed Fee · Retainer · AMS
- SLA configuration per contract (uptime, hours, response time)
- Payment terms: Net 15, Net 30 with monthly billing cycle
- Billing milestones with completion status per Milestone contract
- Quick filter tabs: All · Active · T&M · Milestone · Fixed Fee · Retainer · AMS

REQUIREMENTS

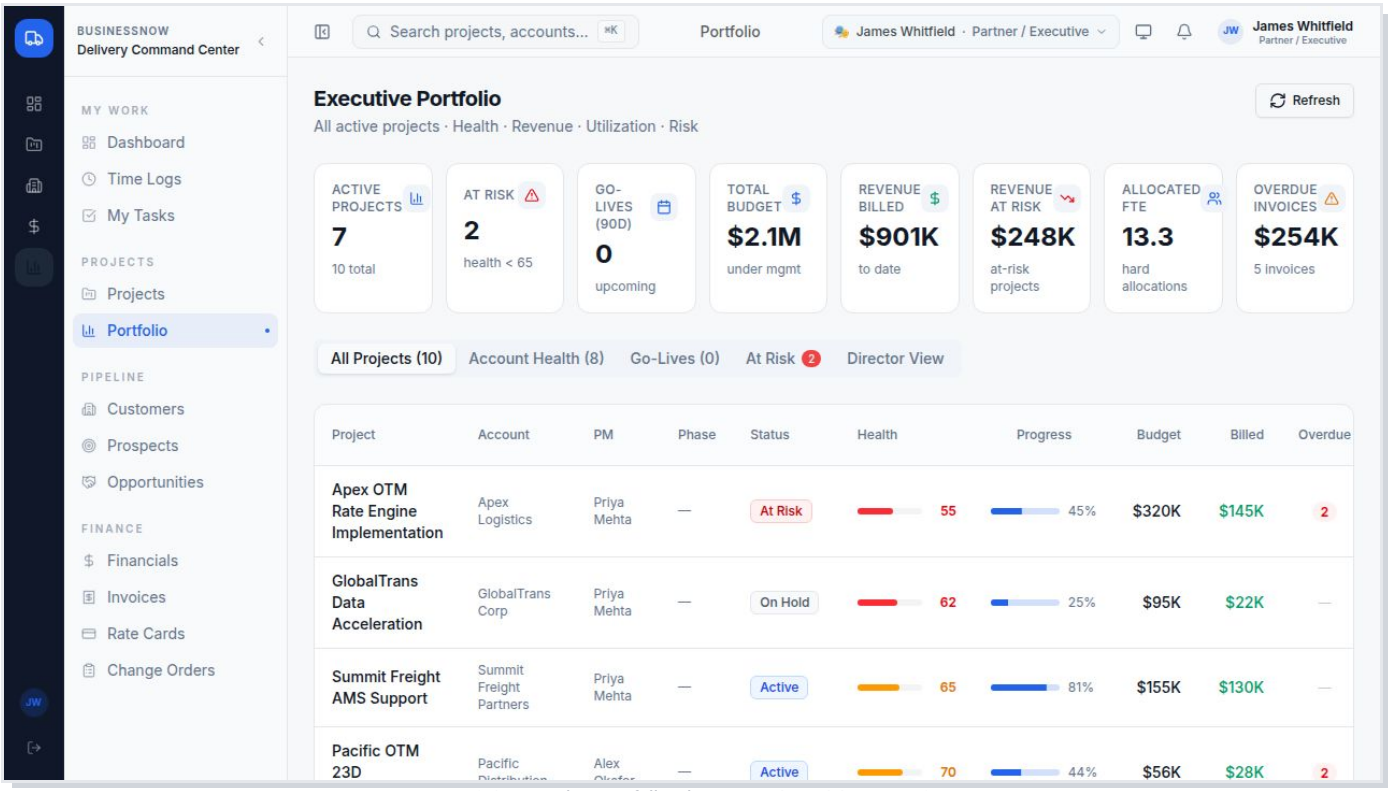
REQ ID			
REQ-CTR-01	TCV, invoiced, and remaining value KPI cards	Must	Finance
REQ-CTR-02	Support all 5 billing models with type badge	Must	Finance
REQ-CTR-03			

SLA configuration and payment terms per contract	Should	Finance, AM

24 Executive Portfolio View

Partner / Executive

Read-only portfolio dashboard for Partners and Executives. 9 projects, $2.1M total budget, $901K billed, 13.3 FTE allocated, 2 at-risk projects with health < 65.



%² Executive Portfolio View — BUSINESSNow v1.0

KEY FEATURES

- 6 active projects out of 9 total in portfolio
- At-risk count: 2 projects with health < 65
- Go-Lives in 90 days: 0 upcoming
- Revenue Billed: $901K, Revenue At Risk: $248K
- Allocated FTE: 13.3 (hard allocations only)
- Tabs: All Projects · Account Health · Go-Lives · At Risk · Director View

REQUIREMENTS

REQ ID			
REQ-PORT-01	Portfolio KPI header: active, at-risk, go-lives, budget	Must	Executive
REQ-PORT-02	Project table with health, PM, budget, billed, overdue	Must	Executive

REQ-PORT-03

At-risk tab filtering projects below health threshold

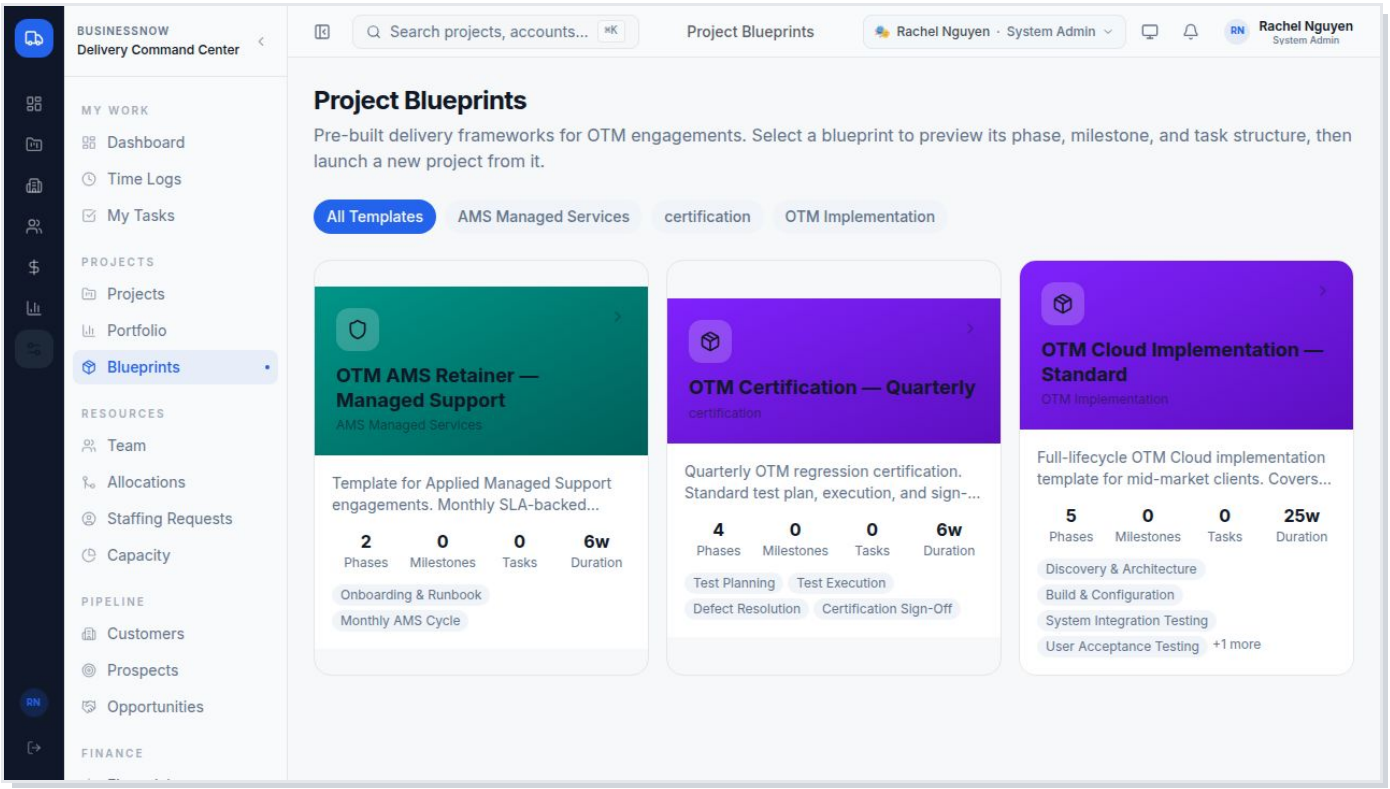
Must

Executive, DD

25 Project Blueprints (Templates)

Admin · PMO Analyst

Pre-built delivery frameworks for OTM engagements. 3 templates: AMS Managed Services, OTM Certification, Cloud Implementation. Used to launch new projects with pre-defined structure.



%² Project Blueprints (Templates) — BUSINESSNow v1.0

KEY FEATURES

- OTM AMS Retainer: 2 phases, 6-week baseline
- OTM Certification — Quarterly: 4 phases, 6-week cycle
- OTM Cloud Implementation: 5 phases, 25-week program
- Phase names, milestone placeholders, task structure included
- Filter by engagement type: AMS, Certification, Implementation
- "Launch Project" action creates new project from blueprint

REQUIREMENTS

REQ ID			
REQ-TMPL-01	Template cards with phase, milestone, task counts	Must	Admin, PMO
REQ-TMPL-02	Filter by engagement type	Must	Admin
REQ-TMPL-03	Launch new project pre-populated from		

blueprint

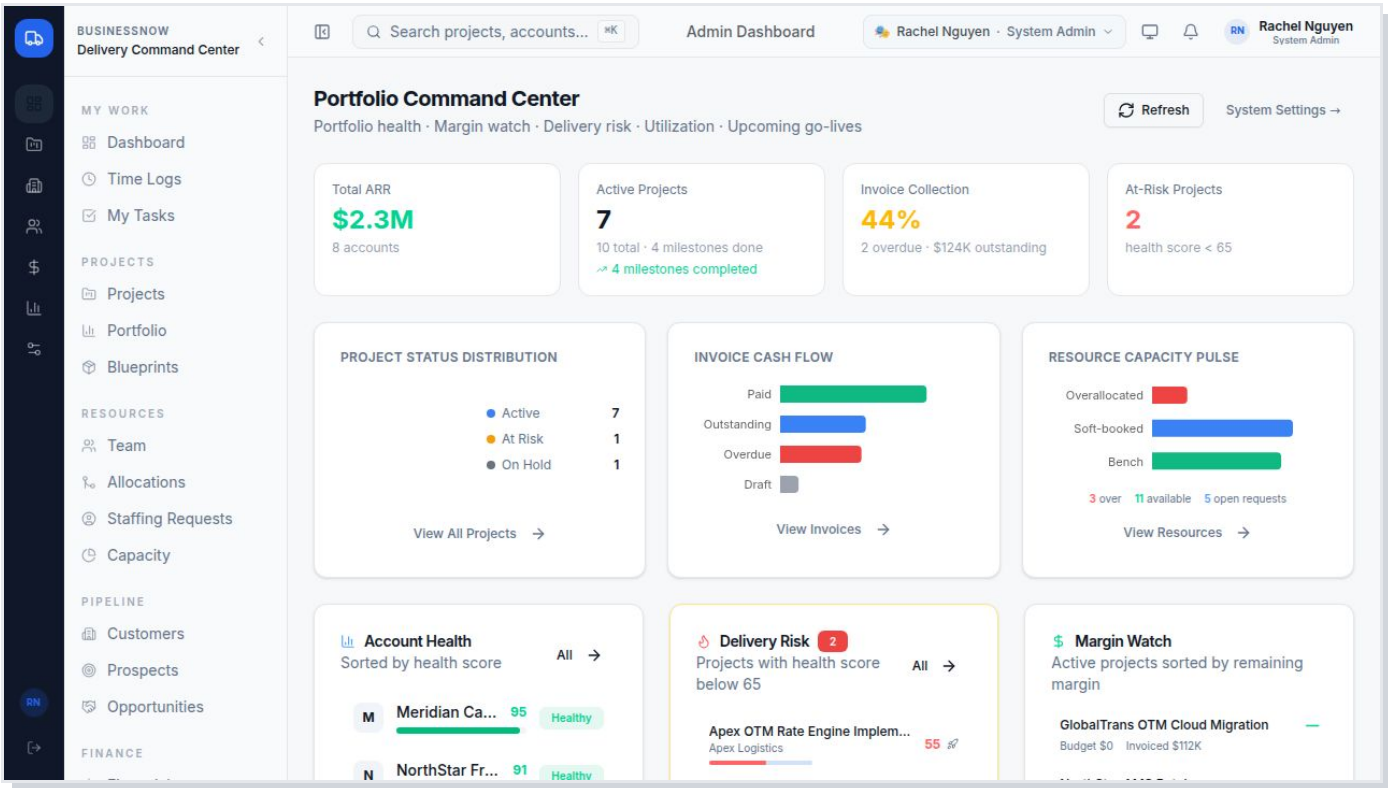
Must

PM, Admin

26 Admin Dashboard — Portfolio Command Center

System Admin

Highest-level operational view for Admins. ARR, active projects, invoice collection rate, at-risk projects, plus charts for project status distribution, invoice cash flow, and resource capacity pulse.



%² Admin Dashboard — Portfolio Command Center — BUSINESSNow v1.0

KEY FEATURES

- Total ARR: $2.3M across 8 accounts
- Invoice Collection: 33% (2 overdue, $124K outstanding)
- Project Status Distribution: 6 Active, 1 At Risk, 1 On Hold
- Resource Capacity Pulse: 3 over, 11 available, 5 open requests
- Account Health sorted by score
- Delivery Risk section: projects with health < 65

REQUIREMENTS

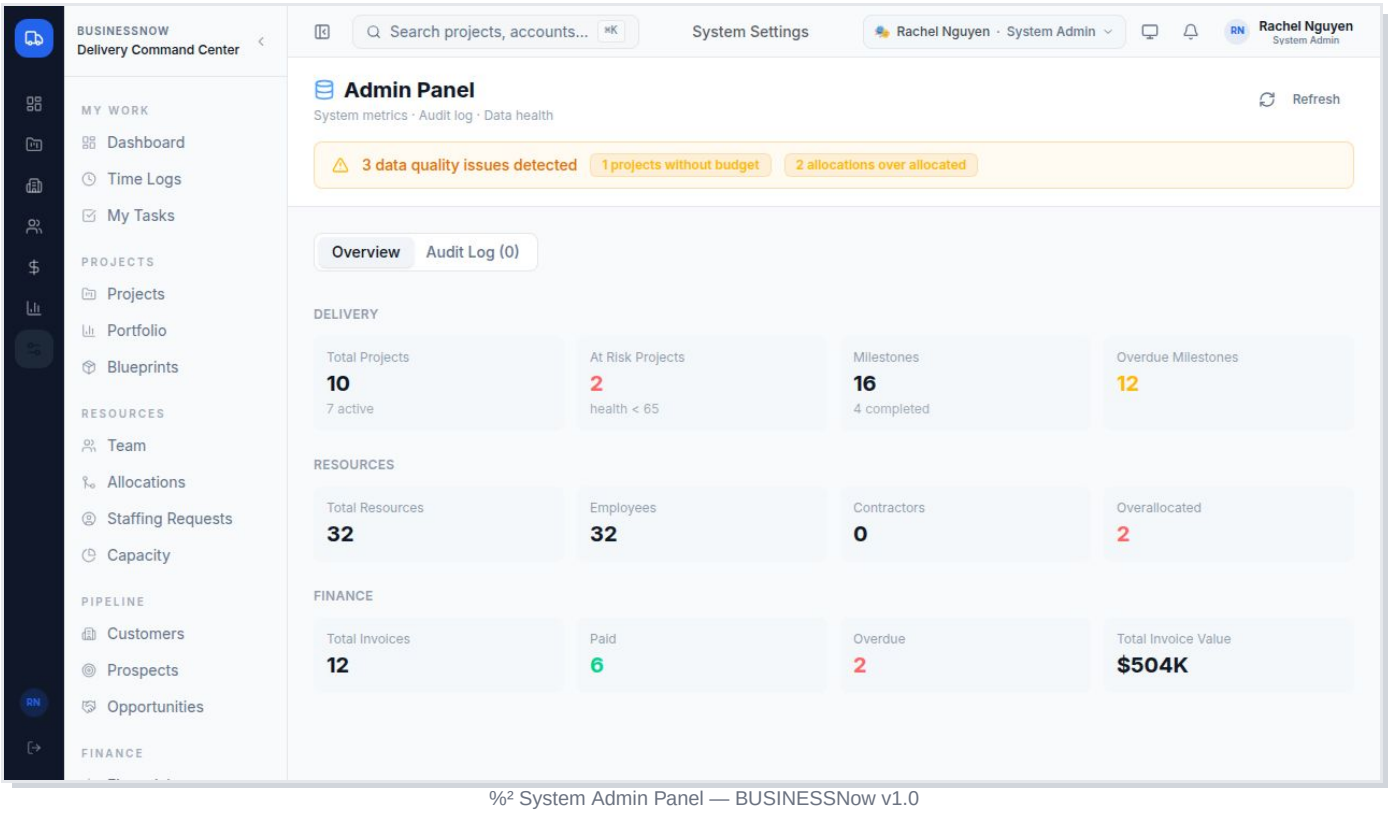
REQ ID			
REQ-ADASH-01	ARR, active projects, collection rate, at-risk KPIs	Must	Admin, DD
REQ-ADASH-02	Project status distribution chart	Must	Admin
REQ-ADASH-03	Resource capacity pulse with availability		

	breakdown	Must	Admin, RM
REQ-ADASH-04			
	Margin Watch sorted by remaining margin	Should	Finance, Admin

27 System Admin Panel

System Admin

Platform governance console. Shows delivery metrics, resource counts, financial summary, entity counts, audit log, and data health checks — all in one admin dashboard.



%² System Admin Panel — BUSINESSNow v1.0

KEY FEATURES

- Delivery: 9 projects, 2 at-risk, 16 milestones, 12 overdue
- Resources: 32 employees, 0 contractors, 2 overallocated
- Finance: 12 invoices, 6 paid, 2 overdue, $672K total
- Tabs: Overview · Audit Log · Data Health
- Data quality issues surfaced (2 over-allocation alerts)
- Entity counts for all 32 database tables

REQUIREMENTS

REQ ID			
REQ-ADMIN-01	Delivery, resource, finance metric overview	Must	Admin
REQ-ADMIN-02	Audit log with timestamp and actor tracking	Must	Admin
REQ-ADMIN-03	Data health checks with issue count and details		

Must

Admin

28 · Database Schema Overview

PostgreSQL + Drizzle ORM — 32 tables

CORE DELIVERY

projects

```
id, customer_id, name, status, health_score, phase, budget_total, budget_spent,
go_live_date, pm_user_id, type, notes, created_at
```

milestones

```
id, project_id, name, status, due_date, payment_amount, payment_trigger,
completed_at
```

tasks

```
id, project_id, milestone_id, name, assignee_id, status, priority, due_date,
estimated_hours, logged_hours, etc_hours
```

project_phases

```
id, project_id, name, order_index, start_date, end_date, status
```

project_updates

```
id, project_id, author_id, content, type, created_at
```

PEOPLE & CAPACITY

resources

```
id, user_id, name, role, seniority, specialization, employment_type,
utilization_target, location, timezone
```

allocations

```
id, resource_id, project_id, role, allocation_pct, hours_per_week, start_date,
end_date, type, status
```

staffing_requests

```
id, project_id, role, skills, allocation_pct, start_date, end_date, priority,
status, fulfilled_by_id
```

resource_skills

```
id, resource_id, skill, experience_years, last_used
```

resource_certifications

id, resource_id, name, issued_by, issued_at, expires_at

TIME & APPROVALS

time_entries

id, resource_id, project_id, task_id, date, hours, billable_hours, notes, status, approved_by, approved_at

timesheet_weeks

id, resource_id, week_start, week_end, status, submitted_at, approved_by_id, approved_at, total_hours

FINANCE

invoices

id, customer_id, project_id, contract_id, invoice_number, amount, status, issued_date, due_date, paid_date, notes

rate_cards

id, name, role, sell_rate, cost_rate, currency, effective_from, effective_to, project_id, is_global

change_orders

id, project_id, name, description, scope_impact, cost_impact, timeline_impact_days, status, submitted_at, approved_at

contracts

id, customer_id, project_id, contract_number, billing_model, total_value, start_date, end_date, payment_terms, billing_cycle, status

contract_slas

id, contract_id, metric, target_value, unit, monitoring_cadence

billing_milestones

id, contract_id, name, amount, status, due_date, invoiced_at

PIPELINE & ACCOUNTS

customers

id, name, industry, segment, status, region, otm_version, acv, renewal_date, health_score, notes

prospects

id, name, industry, segment, region, acv_potential, source, status, owner_id, notes

opportunities

id, customer_id, prospect_id, name, description, type, value, probability, stage, close_date, owner_id

opportunity_activities

id, opportunity_id, type, notes, author_id, created_at

SYSTEM

users

id, email, name, role, avatar_url, is_active, created_at

templates

id, name, type, description, phases_json, created_by, created_at

notifications

id, user_id, type, title, body, read, created_at, entity_type, entity_id

audit_logs

id, user_id, action, entity_type, entity_id, diff_json, created_at

data_health_issues

id, type, severity, entity_type, entity_id, description, detected_at, resolved_at

29 · API Reference

Express 5 REST API — 35+ routes — Port 8080 — Phase 5 Security Hardened

All routes are prefixed with /api. Role-based access is enforced via the X-User-Role header, which is injected by the frontend's fetch interceptor. No routes are publicly accessible without a role header.

PHASE 5 — SECURITY HARDENING

Rate limiting · RBAC middleware · Request logging · Sentry frontend monitoring

Rate Limiting

100 requests / 15-minute window per IP. Returns HTTP 429 when exceeded. Configured via express-rate-limit on all /api/* routes.

Auth Middleware

requireRole(...roles) guard on every route. Reads X-User-Role header; rejects with 401 if missing or 403 if role is not permitted.

Request Logger

morgan 'combined' format logs every request: method, path, status code, response time, and IP address to stdout.

Sentry (Frontend)

@sentry/react captures runtime exceptions, unhandled promise rejections, and React error boundaries in production. DSN configured via VITE_SENTRY_DSN.

RBAC Field Access

Finance fields (bill_rate, cost_rate, margin) filtered server-side based on role — consultants receive null for sensitive financial columns.

/api/projects

GET	`/`	List all projects with health, budget, PM, phase
GET	`/:id`	Get project detail — all 9 tabs' data aggregated
GET	`/:id/team`	Project team members with allocation details

GET	/:id/milestones	Project milestones with status and payment trigger
GET	/:id/tasks	Project task list with assignee and status
GET	/:id/time-entries	Time entries for project (all resources)
GET	/:id/finance	WIP, invoices, margin, contract for project

/api/resources

GET	/	All resources with utilization heatmap data
GET	/:id	Resource profile, skills, certifications, 12-wk load
GET	/capacity	12-week capacity forecast by specialization

/api/allocations

GET	/	All allocations — hard & soft with conflict detection
POST	/	Create new allocation (hard or soft)
PUT	/:id	Update allocation percentage or dates

DELETE	/:id	Remove allocation

/api/finance

GET	/	WIP value, receivables, overdue invoices, TCV
GET	/invoices	All invoices with status and aging
GET	/rate-cards	Global and project-specific rate cards
GET	/change-orders	All change orders across projects
GET	/contracts	All contracts with SLA and billing milestones

/api/timesheets

GET	/my	Current user's time entries (week filter)
POST	/	Log a time entry for a project/task
GET	/approval	All submitted timesheets pending PM approval
POST	/approve	Approve or reject a timesheet entry

/api/customers & /api/pipeline

GET	`/api/customers`	All customers with health, ACV, OTM version
GET	`/api/customers/:id`	Customer health breakdown, invoice summary, projects
GET	`/api/opportunities`	Opportunity pipeline with stage and value
GET	`/api/staffing-requests`	Open staffing requests across all projects
GET	`/api/dashboard/admin`	Admin portfolio command center data
GET	`/api/dashboard/pm`	PM delivery workspace — urgent items

30 · End-to-End Process Flow

6-stage delivery lifecycle managed in BUSINESSNow

1 PIPELINE

- Prospect entered in Customers module with status=Prospect
- Opportunity created and tracked through sales pipeline
- SOW and rate card negotiated, contract created
- Deal closed — account converted to active Customer

2 INITIATION

- Project created from Blueprint template
- PM assigned, go-live date and budget set
- Team staffed via Staffing Requests !' Allocations
- Contract attached, billing model configured

3 PLANNING

- Phases and milestones defined from blueprint
- Tasks created with assignees, estimates, due dates
- Baseline budget set for burn tracking
- Rate cards applied to project for WIP calculation

4 EXECUTION

Consultants log time weekly via
Timesheet grid

PMs review and approve/reject time entries

Milestones marked complete, triggering payment

Change orders raised for scope changes

5 FINANCE

- WIP engine calculates billable value from approved hours
- Finance creates invoices against milestones / WIP
- Invoices sent to client, tracked in AR aging
- Payments collected, margin % tracked vs target

6 CLOSE

- All milestones completed, project closed
- Final invoice issued and collected
- Resource allocations removed, capacity freed
- Project archived, health history retained for AM

BUSINESSNow

Delivery Command Center

Built for KSAP Technologies
Oracle Transportation Management Consulting

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